

MAS2001 Probability and Statistics

Illustrated study notes

Eleven audited Markdown notes with topic diagrams, exact formulas, worked examples, and source-preserving code panels.

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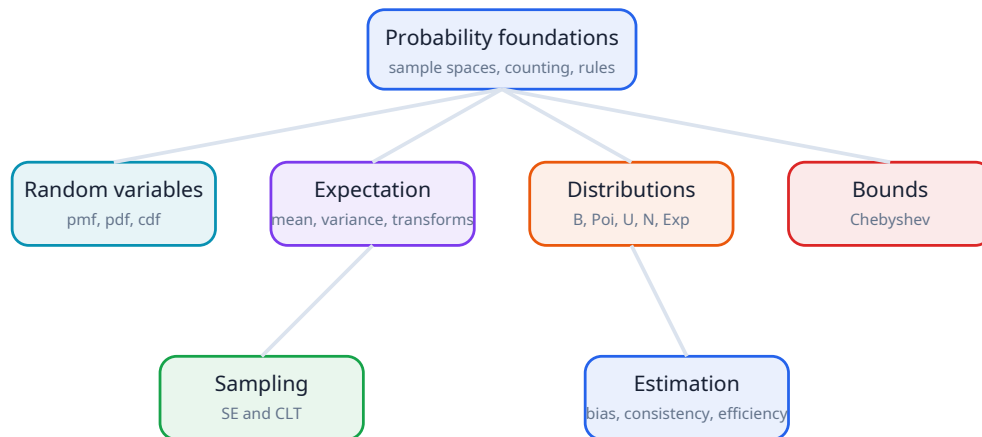
- 00. NOTES: the whole course, written out
- 01. 01 PROBABILITY FOUNDATIONS (lecture 2, notes p012-p050)
- 02. 02 RANDOM VARIABLES (notes p051-p062)
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NOTES: the whole course, written out

These are real notes. Every slide is rewritten as something you can read and learn from without opening a PDF. Worked examples keep their exact numbers. Original ASCII diagrams are retained as source evidence, with vector figures inserted beside the relevant explanations.

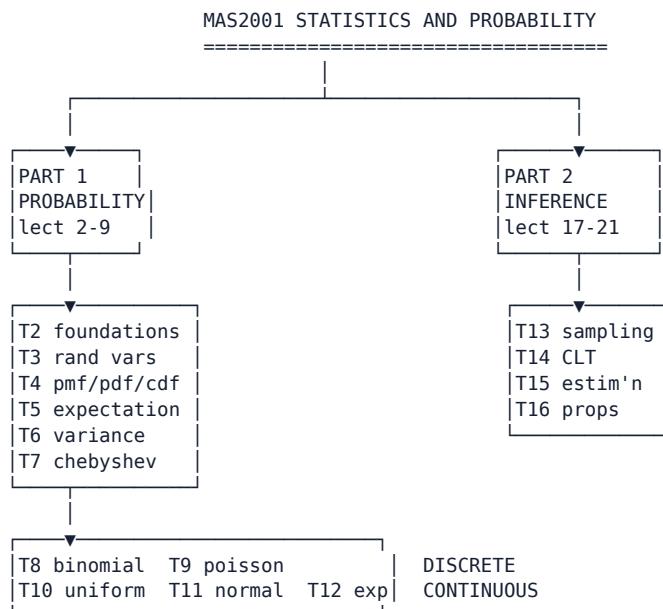
The map

MAS2001 MTE concept map



Overview only. Read connectors as prerequisites, not as claims of independence.

Course concept map



File list

file	covers	lecture
00-NOTES-INDEX.md	this map	-
01-probability-foundations.md	lecture 2, notes p012-p050	2

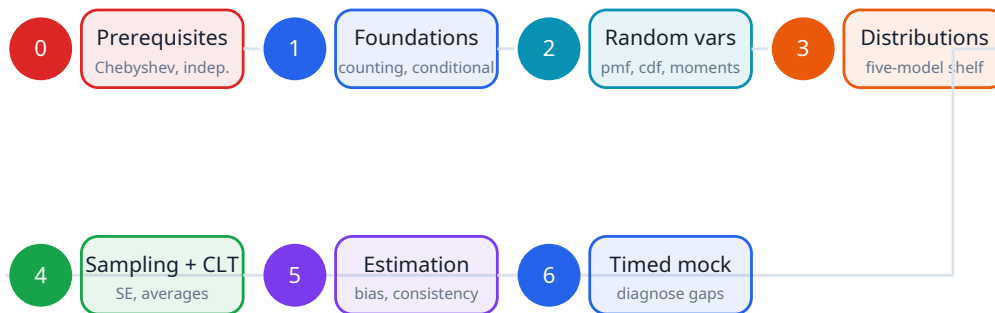
02-random-variables.md	notes p051-p062	3-4
03-pmf-and-cdf.md	notes p063-p087	5-6
04-expectation-and-variance.md	notes p088-p111	7-9
05-continuous-rv.md	notes p112-p147	5-6 cont
06-binomial-poisson.md	ppt3 (28 pages)	12-13
07-uniform-normal-exponential.md	ppt4 (44 pages)	14-16
08-sampling-and-clt.md	lms-standard-error-clt (19 pages)	17-18
09-estimation.md	ppt5 (26 pages) + lms-theory	19-21
10-chebyshev-and-hidden.md	Chebyshev deck + H1-H8	10-11

How the numbers in these notes were handled

The slides have the errata listed in `../08-TRAPS.md`. In these notes I write the CORRECT value and mark the slide's wrong value inline with `[slide prints X]`. That way the note is right and you still know what the slide said.

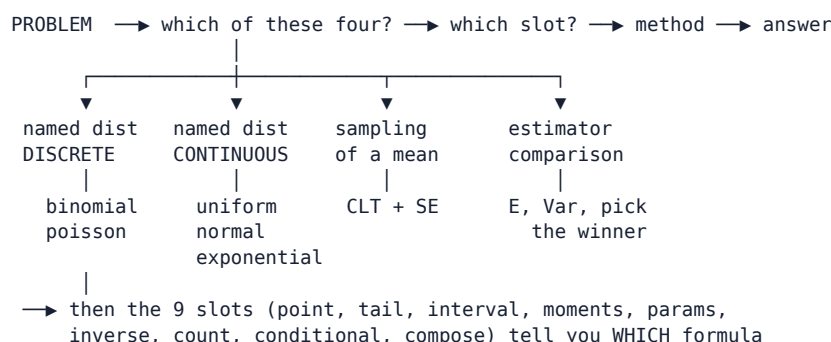
The one picture to hold in your head for the whole course

Study order with dependency gates



advance only after the gate for the current pass is met

Study order and dependency gates



01 PROBABILITY FOUNDATIONS (lecture 2, notes p012-p050)

1. What statistics is (p012-p016) - not examined, but the framing

statistics = the science of COLLECTION, PRESENTATION, ANALYSIS, INTERPRETATION of numerical data

"All statistics are numerical statements, but not all numerical statements are statistics."

Ram has 100 Rs in his pocket -> NOT statistics (one fact, no data set)
avg age of 5th class is 10 yrs -> IS statistics (a derived numerical statement)

Four limitations the slides list (worth a one-line recall, low exam weight): 1. not suited to qualitative phenomena 2. does not study individuals 3. statistical laws are not exact 4. liable to be misused (the customer-satisfaction survey only of loyal customers)

2. Probability, the intuition (p018-p019)

P near 1 = event extremely likely

$P = 0.5$ = a coin-flip situation

P near 0 = rare event

$0 \leq P(E) \leq 1$ ALWAYS

0	0.5	1
impossible	even	certain
rare	coin flip	common

3. Random experiment (p023)

DEFINITION

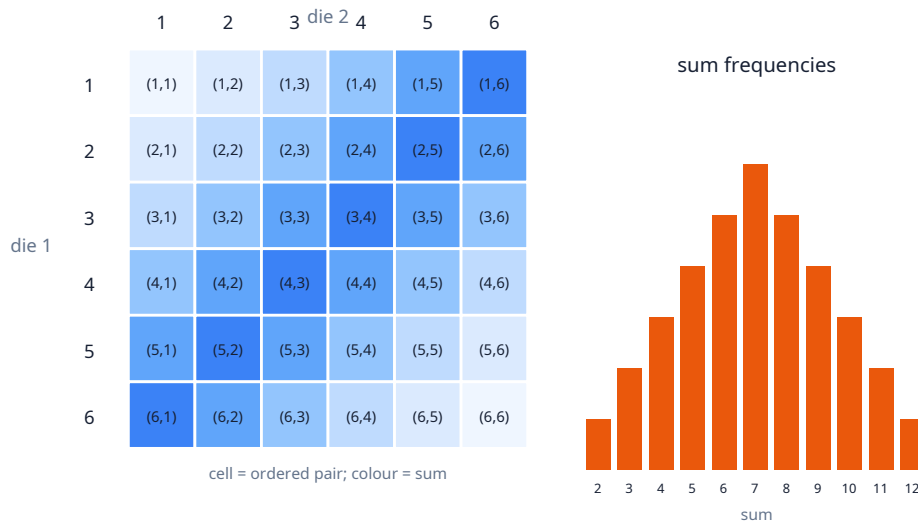
an experiment whose outcome cannot be predicted with certainty in advance, although ALL possible outcomes are known

EXPERIMENT	POSSIBLE OUTCOMES
toss a coin	H, T
roll a die	1,2,3,4,5,6
draw a card	any one of 52
pick a student	any student in the class

note: we KNOW all outcomes, we do not know WHICH ONE occurs

4. Sample space S (p024)

Two dice: 36 ordered outcomes



The 36 ordered outcomes for two dice

S = set of ALL possible outcomes

one coin	S = {H, T}	2 outcomes
one die	S = {1,2,3,4,5,6}	6 outcomes
two coins	S = {HH, HT, TH, TT}	4 outcomes
two dice	S = {(1,1), (1,2), ..., (6,6)}	6 x 6 = 36 outcomes

THE COUNTING RULE THAT SAVES MARKS
 two dice are counted as 36 ORDERED pairs, not 21 unordered
 (1,2) and (2,1) are DIFFERENT outcomes
 every dice probability (1/36, 2/36, 3/36...) assumes 36

The 36 ordered-pair positions, with each cell displaying its sum:

	1	2	3	4	5	6	<- die 2
1	2	3	4	5	6	7	
2	3	4	5	6	7	8	
3	4	5	6	7	8	9	
4	5	6	7	8	9	10	
5	6	7	8	9	10	11	
6	7	8	9	10	11	12	

cell position = (row die, column die)
 printed value = the sum; each cell = 1/36

die 1

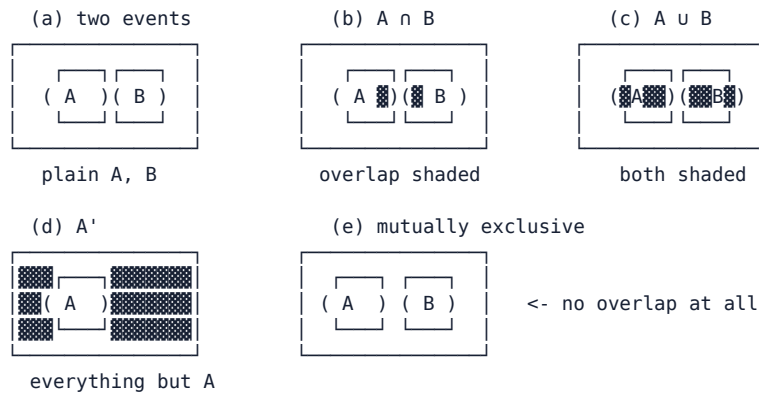
count the 7s: 6 of them -> $P(\text{sum}=7) = 6/36 = 1/6$ (the most likely sum)
 count the 2s: 1 (top-left) -> $1/36$
 count the 12s: 1 (bottom-right) -> $1/36$
 the counts run 1,2,3,4,5,6,5,4,3,2,1 (sums 2..12) = the familiar triangle

5. Events and set algebra (p030-p040)

outcome	a single possible result
event	a SUBSET of S (a collection of outcomes)
null event	the empty set, no outcomes

A'	complement	"not A"	everything in S not in A
$A \cup B$	union	"A or B"	in A, or B, or BOTH
$A \cap B$	intersection	"A and B"	in BOTH
disjoint / mutually exclusive : $A \cap B = \emptyset$			

ASCII Venn diagrams:



Demorgan, drawn as the two rules the slides state:

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

"neither" = not-A and not-B "not both" = not-A or not-B

6. Mutually exclusive vs independent (p025-p028) - THE classic trap

MUTUALLY EXCLUSIVE

cannot happen together
 $A \cap B = \emptyset$
 $P(A \cup B) = P(A) + P(B)$

INDEPENDENT

one happening does not change
the chance of the other
 $P(A \cap B) = P(A) \times P(B)$
or $P(A|B) = P(A)$

THEY ARE NOT THE SAME, AND THEY ARE NOT OPPOSITES

exclusive -> CANNOT both happen -> knowing A KILLS B
(so NOT independent,
when $P > 0$)

Worked: the die counter-example (p025)

$A = \text{even} = \{2, 4, 6\}$ $B = \text{prime} = \{2, 3, 5\}$
 $A \cap B = \{2\}$ -> they SHARE the outcome 2
-> NOT mutually exclusive

$A = \text{even} = \{2, 4, 6\}$ $B = \text{odd} = \{1, 3, 5\}$
 $A \cap B = \emptyset$ -> mutually exclusive

Worked: coin + die independence (p027)

$A = \text{head}, P(A) = 1/2$
 $B = 6, P(B) = 1/6$
 $P(A \cap B) = 1/2 \times 1/6 = 1/12$ -> independent

Worked: draw + replace (p027) -> independent. draw, no replace -> dependent.

Mutual independence of n events (p028): for EVERY subset of size k ($k = 2..n$),

$$P(A_{i1} \cap A_{i2} \cap \dots \cap A_{ik}) = P(A_{i1}) P(A_{i2}) \dots P(A_{ik})$$

7. Counting: the engine of half the course (p043)

ORDER MATTERS ?

ORDER DOES NOT MATTER ?

PERMUTATION

$P(n,r) = n!/(n-r)!$
 "arrangements"

COMBINATION

$C(n,r) = n!/(r!(n-r)!)$
 "choose", "committee", "sets of"

with REPEATED items:

$n! / (n_1! n_2! \dots n_k!)$ $n_1, n_2, \dots$ are the repeat counts

WORD TRIGGERS

"arrange", "lineup", "order", "rank" -> PERMUTATION
 "choose", "select", "committee", "set" -> COMBINATION

and $C(n,r) = C(n,n-r)$ (choosing r to keep = choosing $n-r$ to drop)

8. The axioms and rules (p044-p045)

AXIOMS

$P(S) = 1$

$P(A) \geq 0$

disjoint $A_1, A_2, \dots$: $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

RULES

addition $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

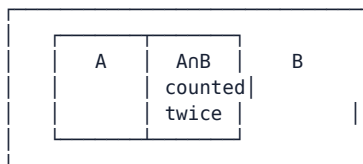
disjoint case $P(A \cup B) = P(A) + P(B)$

multiplication $P(A \cap B) = P(A) P(B|A) = P(B) P(A|B)$

independent $P(A \cap B) = P(A) P(B)$

complement $P(A') = 1 - P(A)$

The addition-rule picture (why you subtract the overlap):



$P(A) + P(B)$ counts AnB TWICE
 subtract it ONCE

9. Worked example 1: tournament (p041-p042)

Four universities 1,2,3,4. Round 1: 1v2, 3v4. Winners play for the title, losers play too.

one outcome is written as a 4-letter code, e.g. 1324:

1 beats 2, 3 beats 4 (round 1) then 1 beats 3, 2 beats 4

$S = \{ 1324, 1342, 1423, 1432, 2314, 2341, 2413, 2431, 3124, 3142, 4123, 4132, 3214, 3241, 4213, 4231 \}$ -> 16 outcomes

(b) $A = 1$ wins the tournament -> 1 must be FIRST in the code
 $A = \{1324, 1342, 1423, 1432\}$ 4 outcomes

(c) $B = 2$ gets into the final -> 2 must be 1st or 2nd
 $B = \{2314, 2341, 2413, 2431, 3214, 3241, 4213, 4231\}$ 8 outcomes

(d) $A \cup B = 12$ outcomes
 $A \cap B = \emptyset$ because 1 and 2 CANNOT both reach the final
 $A' = S - A = 12$ outcomes

10. Worked example 2: faculty laptops (p046-p047)

6 faculty, 2 chose laptop, 4 chose desktop. Only 2 setups per day, the 2 are chosen randomly.

total outcomes = $C(6,2) = 15$ (choose 2 of the 6 computers)

(a) both laptops = $C(2,2)/15 = 1/15$

(b) both desktops = $C(4,2)/15 = 6/15 = 2/5$

(c) at least one desktop : complement is "both laptops" = A
 $P(C) = 1 - P(A) = 1 - 1/15 = 14/15$

(d) at least one of each type:

$$1 - P(A \cup B) = 1 - 1/15 - 2/5 = 1 - 1/15 - 6/15 = 8/15$$

(A and B are disjoint, so no overlap to subtract)

11. Worked example 3: basketball lineup (p048-p049)

STARTING LINEUP = 2 guards + 2 forwards + 1 center

Part (a): 3 centers, 4 guards, 4 forwards, 1 swing player X

no X	$C(4,2) \times C(4,2) \times C(3,1) = 6 \times 6 \times 3 = 108$
X as guard	$C(4,1) \times C(4,2) \times C(3,1) = 4 \times 6 \times 3 = 72$
X as forward	$C(4,2) \times C(4,1) \times C(3,1) = 6 \times 4 \times 3 = 72$
	TOTAL = 252 lineups

Part (b): 5 guards, 5 forwards, 3 centers, 2 swing (X and Y)
choose 5 of the 15 at random; P(valid lineup)?

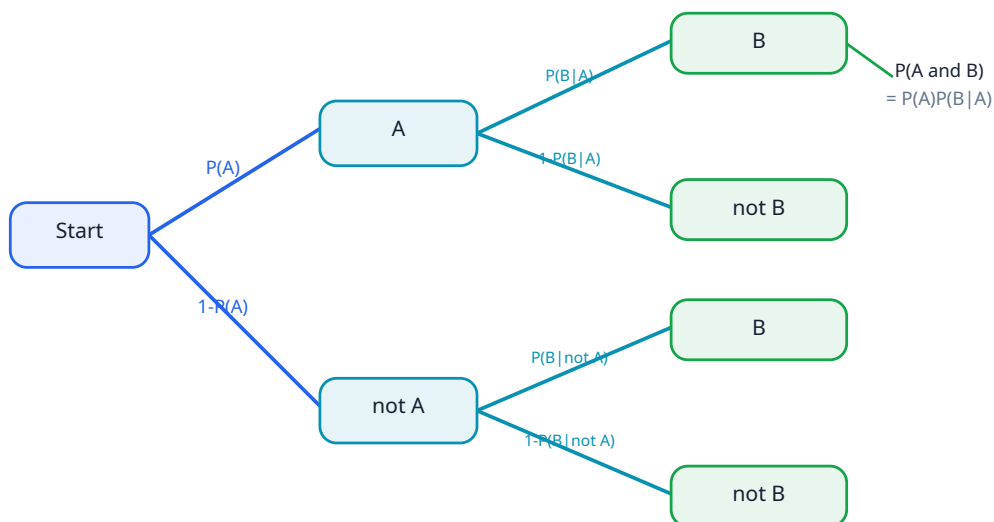
neither X nor Y	$C(5,2)C(5,2)C(3,1) = 300$
X only	$C(5,1)C(5,2)C(3,1) \times 2 \text{ cases} = 300$
Y only	same = 300
both X and Y	$C(5,1)C(5,1)C(3,1) + 2C(5,2)C(3,1) = 135$
	TOTAL = 1035

$$P = 1035 / C(15,5) = 1035/3003$$

THE METHOD: split into disjoint cases by WHO of the swing players is used, count each case, ADD. Never double count.
no X, no Y + X only + Y only + both X and Y = total

12. Conditional probability (p050)

Conditional probability tree



Conditional probability tree

$$P(A|B) = P(A \cap B) / P(B), \quad P(B) > 0$$

Worked: digital camera buyers

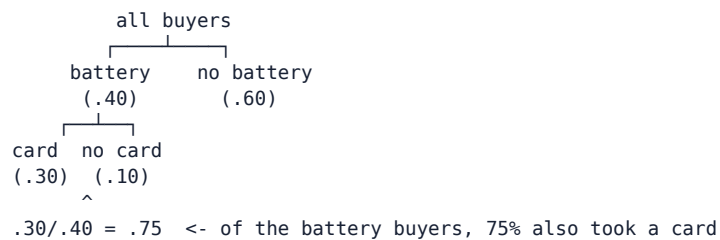
60% buy a memory card	$P(A) = .60$
40% buy an extra battery	$P(B) = .40$
30% buy both	$P(A \cap B) = .30$

$$P(\text{card} \mid \text{battery}) = .30/.40 = .75$$

$$P(\text{battery} \mid \text{card}) = .30/.60 = .50$$

NOTE THE TWO ANSWERS ARE DIFFERENT. .75 is NOT .50.
 The denominator is WHAT YOU ARE CONDITIONING ON.
 $P(A|B)$ and $P(B|A)$ are different numbers.

The tree that makes it obvious:



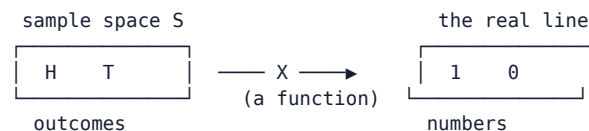
That is lecture 2 in full. The two things to carry forward: (1) the 36-cell dice grid, (2) exclusive is not independent.

02 RANDOM VARIABLES (notes p051-p062)

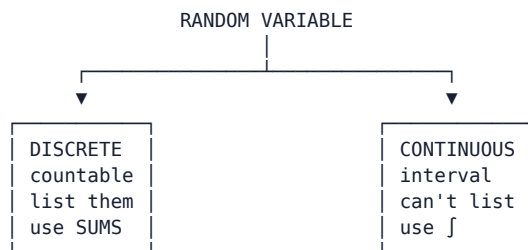
1. What a random variable IS (p051)

A random variable is a FUNCTION whose domain is the sample space and whose range is the set of real numbers.

It is NOT the outcome. It is the NUMBER you attach to it.



$$X(H) = 1 \quad X(T) = 0$$



2. Example 1: Bernoulli (p052)

toss one fair coin, $S = \{H, T\}$
define $X(H) = 1$, $X(T) = 0$

DEFINITION: any rv whose only values are 0 and 1 is a BERNOLLI random variable.

Bernoulli is the atom of the binomial: one trial, success = 1, failure = 0.

3. Example 2: two coins (p053)

$S = \{HH, HT, TH, TT\}$
 $X = \text{number of heads}$

outcome	X value
HH	2
HT	1
TH	1
TT	0

$$X(HH)=2, \quad X(TT)=0, \quad X(TH)=1, \quad X(HT)=1$$

Notice HT and TH give the SAME X . Different outcomes, same value. That collapse is how a pmf is born (next file).

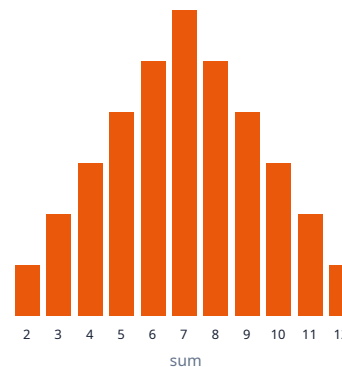
4. Example 3: dice sum (p054-p055)

Two dice: 36 ordered outcomes

	1	2	3	4	5	6
die 1						
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

cell = ordered pair; colour = sum

sum frequencies

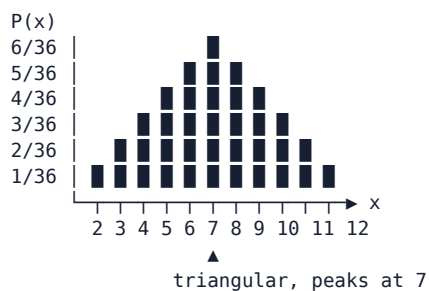


Ordered outcomes and the distribution of their sum

X = sum of two fair dice

x	outcomes producing it	count	P(X=x)
2	(1,1)	1	1/36
3	(1,2) (2,1)	2	2/36
4	(1,3) (2,2) (3,1)	3	3/36
5	(1,4) (2,3) (3,2) (4,1)	4	4/36
6	5 outcomes	5	5/36
7	6 outcomes	6	6/36
8	5 outcomes	5	5/36
9	4 outcomes	4	4/36
10	3 outcomes	3	3/36
11	(5,6) (6,5)	2	2/36
12	(6,6)	1	1/36
	Σ	36/36 = 1	✓

The pmf as a picture:



The identity the slide writes:

$$1 = P\left(\bigcup_{n=2}^{12} (X=n)\right) = \sum_{n=2}^{12} P(X=n)$$

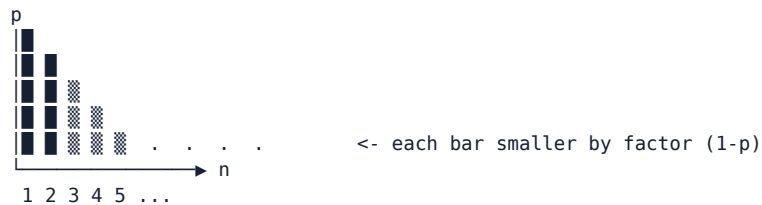
5. Example 4: two coins as Y = heads (p056)

$$\begin{aligned}
 P(Y=0) &= P(TT) &= 1/4 \\
 P(Y=1) &= P(HT, TH) &= 2/4 \\
 P(Y=2) &= P(HH) &= 1/4 \\
 \text{sum} &= 1 \quad \checkmark
 \end{aligned}$$

6. Example 5: geometric (p057-p059) - counts flips until first head

N = number of flips needed to get the first head

$$\begin{aligned} P(N=1) &= P(H) &&= p \\ P(N=2) &= P(TH) &&= (1-p)p \\ P(N=3) &= P(TTH) &&= (1-p)^2 p \\ \dots &&& \\ P(N=n) &= P(T\dots TH) &&= (1-p)^{n-1} p \end{aligned}$$



The sum verification (infinite geometric series, common ratio $(1-p)$):

$$\sum (1-p)^{n-1} p = p / (1 - (1-p)) = p/p = 1 \quad \checkmark$$

This is the GEOMETRIC distribution. Mean = $1/p$.
It has NO teaching slide for its mean, but the exam asks it (assignment 1 short Q4: $E = 2$).

7. Example 6: car lifetime (p060) - the bridge to continuous

X = lifetime of a car, assumed to take any value in an interval (a,b)

- not countable, cannot be listed as a sequence

-> NOT a discrete rv

8. Discrete vs continuous, the definitions (p061-p063)

DISCRETE	CONTINUOUS
finite OR countably infinite can be arranged as a sequence 1st, 2nd, 3rd, ...	all values in an interval CANNOT be listed as a sequence $P(X = \text{any single } c) = 0$

The formal continuous definition has TWO parts:

- possible values = all numbers in one interval (or a union of intervals)
e.g. $[0,10] \cup [20,30]$
- NO single value has positive probability: $P(X = c) = 0$

WHY $P(X=c)=0$ FOR CONTINUOUS
 $P(X=c) = \int \text{from } c \text{ to } c \text{ of } f(x) \, dx = 0$
 the width is zero, an area of zero width is zero
 -> endpoints never matter: $P(a < X < b) = P(a \leq X \leq b)$

9. The summary table to memorise

	DISCRETE	CONTINUOUS
support	list of values	interval(s)
tool	Σ (sum)	$\int$ (integral)
$P(X=x)$	$p(x) > 0$	0 always
$E(X)$	$\Sigma x \cdot p(x)$	$\int x \cdot f(x) \, dx$
$\text{Var}(X)$	$\Sigma (x-\mu)^2 p(x)$	$\int (x-\mu)^2 f(x) \, dx$

named examples	binomial, Poisson, geo	uniform, normal, exponential
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Next file builds the pmf and cdf (the two functions that describe any rv).

03 PMF AND CDF (notes p063-p087)

1. The probability mass function, pmf (p064)

$p(x) = P(X = x) = P(\{s \text{ in } S : X(s) = x\})$
 "for every possible value x , the probability of observing it"

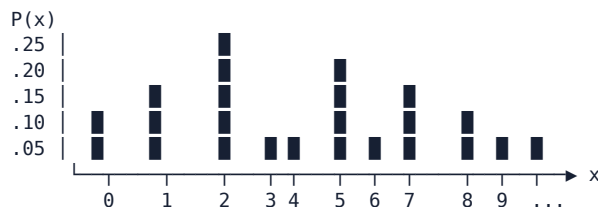
Two conditions (necessary AND sufficient):

- (i) $p(x) \geq 0$ every probability is non-negative
- (ii) $\sum p(x) = 1$ the probabilities over ALL possible x sum to 1

"The probability mass function of a discrete random variable describes the behaviour of the variable."

2. Worked: gas station pumps (p065)

x	0	1	2	3	4	5	6	
$p(x)$	.05	.10	.15	.25	.20	.15	.10	sum = 1.00 ✓



- (a) at most 2 $P(X \leq 2) = .05 + .10 + .15 = .30$
- (b) at least 3 $P(X \geq 3) = 1 - P(X \leq 2) = 1 - .30 = .70$
 or $.25 + .20 + .15 + .10 = .70$
- (c) between 2 and 5 inclusive $P(2 \leq X \leq 5) = .15 + .25 + .20 + .15 = .75$

THE cdf-arithmetic habit:

- "at most k " = $F(k)$ add up to k
- "at least k " = $1 - F(k-1)$ for an integer-valued X
- "between a, b " = $F(b) - F(a-1)$ for integer-valued X and integer a, b

3. Worked: the find-k table (p066) - the batch-1 assignment long Q2

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) FIND k

$$\begin{aligned} \sum p(x) &= k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k \\ &= 9k + 10k^2 \\ 10k^2 + 9k &= 1 \rightarrow 10k^2 + 9k - 1 = 0 \\ 10k^2 + 10k - k - 1 &= 0 \rightarrow (k+1)(10k-1) = 0 \\ k &= 1/10 \quad (\text{or } -1, \text{ rejected: probability cannot be negative}) \end{aligned}$$

(ii) with $k = 1/10$:

$$\begin{aligned} P(X < 6) &= p(0) + p(1) + p(2) + p(3) + p(4) + p(5) \\ &= 0 + k + 2k + 2k + 3k + k^2 \\ &= 8k + k^2 \\ &= 0.80 + 0.01 = 0.81 \\ P(X \geq 6) &= 1 - 0.81 = 0.19 \\ P(0 < X < 5) &= p(1) + p(2) + p(3) + p(4) = k + 2k + 2k + 3k = 8k = 0.80 \end{aligned}$$

quick consistency check: $P(X < 6) + P(X \geq 6) = 0.81 + 0.19 = 1.00$ ✓

(iii) DISTRIBUTION FUNCTION: accumulate

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.1 & 1 \leq x < 2 \end{cases}$$

0.3	$2 \leq x < 3$
0.5	$3 \leq x < 4$
0.8	$4 \leq x < 5$
0.81	$5 \leq x < 6$
0.83	$6 \leq x < 7$
1.0	$x \geq 7$

(iv) minimum a with $F(a) > 1/2$:

$$F(4) = 0.8 > 0.5 \rightarrow a = 4$$

($F(3) = 0.5$ is NOT > 0.5 , strict inequality)

WATCH THE $P(x)=0$ ROW

The batch-1 edition has $P(X=0)=0$, giving $10k^2+9k=1 \rightarrow k=1/10$.

The SECOND edition changed $P(X=0)$ to k , giving $12k+10k^2=1$

$\rightarrow k=0.0782$, but its own key still prints $k=1/10$ which does

NOT satisfy its own table. The second edition is broken

(errata 5.1). Study the batch-1 edition.

4. Worked: tune-ups, cylinders (p067-p069)

45% four-cylinder, 40% six-cylinder, 15% eight-cylinder

X = number of cylinders of the next car

x	4	6	8	
$p(x)$	.45	.40	.15	sum 1.00 ✓

histogram picture:



$$P(X \geq 6) = .40 + .15 = .55$$

$$P(X > 6) = .15 \quad \leftarrow \text{the boundary moved}$$

5. Worked: building permits, proportional pmf (p070-p071)

Y = number of forms required, $p(y) = k y$, $y = 1, \dots, 5$

$$(a) \sum p(y) = k(1+2+3+4+5) = 15k = 1 \rightarrow k = 1/15$$

$$(b) P(Y \leq 3) = (1+2+3)/15 = 6/15 = 0.4$$

$$(c) P(2 \leq Y \leq 4) = (2+3+4)/15 = 9/15 = 0.6$$

$$(d) \text{ is } p(y) = y^2/50 \text{ a pmf? } \Sigma = (1+4+9+16+25)/50 = 55/50 \neq 1 \rightarrow \text{NO}$$

THE VALID-PMF TEST IS ALWAYS: sum over the support = 1 ?
If it does not sum to 1, it is not a pmf. Period.

6. Worked: flashlight batteries (p072-p073) - a pmf built from a process

90% of batteries acceptable (A), need 2 acceptable to work.

Y = number of batteries tested.

$$(a) p(2) = P(AA) = (.9)(.9) = .81$$

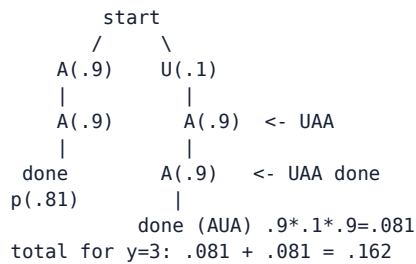
$$(b) p(3) = P(UAA \text{ or } AUA) = (.1)(.9)(.9) + (.9)(.1)(.9) = 2(.1)(.9)^2 = .162$$

$$(c) p(5): \text{ the 5th must be A, exactly one of the first four also A} \\ AUUUA, UAUUA, UUAUA, UUUAA = 4 \text{ outcomes}$$

$$p(5) = 4(.1)^3(.9)^2 = .00324$$

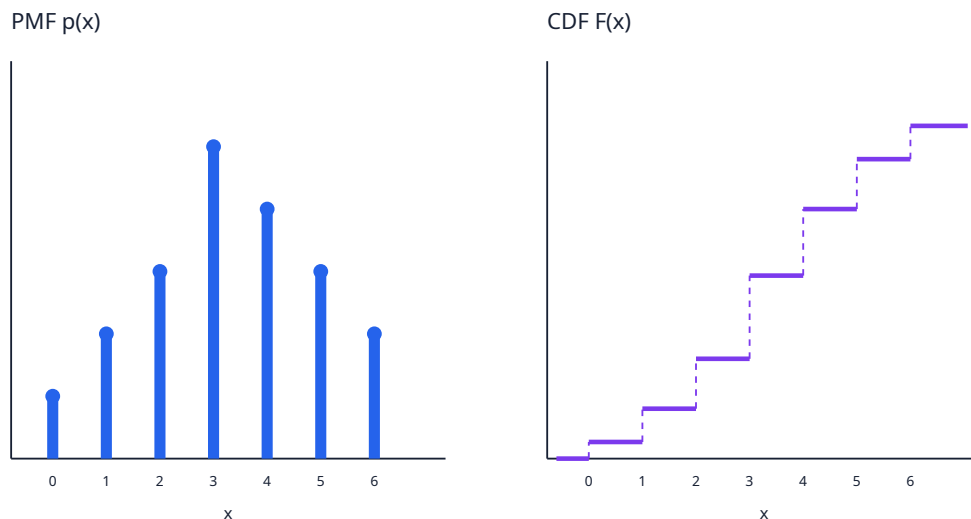
$$(d) \text{ general: } P(Y=y) = (y-1)(.1)^{y-2}(.9)^2, \quad y = 2, 3, 4, \dots$$

The tree for part (b):



7. The cumulative distribution function, cdf (p074-p078)

PMF accumulates into a CDF



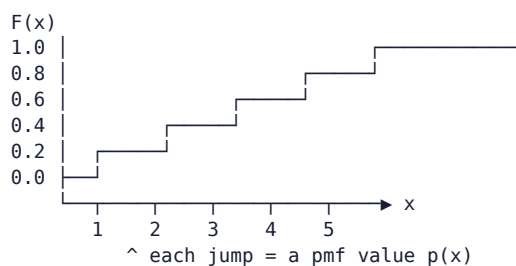
How a PMF accumulates into a CDF

$F(x) = P(X \leq x) = \sum_{\{y: y \leq x\}} p(y)$	(discrete)
$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$	(continuous)

PROPERTIES OF F

non-decreasing (it only accumulates)
 $0 \leq F(x) \leq 1$
 $F(-\infty) = 0, \quad F(+\infty) = 1$
 right-continuous

THE STEP-FUNCTION PICTURE (discrete cdf)



pmf AND cdf DETERMINE EACH OTHER

discrete: $p(x) = F(x) - F(x-)$ (the JUMP at x)

integer rv: $p(n) = F(n) - F(n-1)$

continuous: $f(x) = F'(x)$ (derivative)

8. THE key cdf theorem (p079)

for $a < b$:

$$P(a < X \leq b) = F(b) - F(a)$$

and the integer-valued version that catches people (p135):

$$P(a \leq X \leq b) = F(b) - F(a-1)$$

when X is integer-valued and a, b are integers

^^

the $a-1$ because $F(a-1)$ INCLUDES the value $a-1$, and

$P(a \leq X)$ must EXCLUDE everything below a

picture:

$F(b)$ = area/include everything up to and INCLUDING b

$F(a)$ = everything up to and including a <- but $P(a < X)$ excludes a

$F(a-1)$ = everything up to $a-1$ <- so this is the right one

...	a-1		a		...		b
drop this			keep	keep			

9. Worked: boards inspection (p080-p081)

Lots of 5 boards, 2 inspected. Boards 1,2 are the only defectives.

X = number of defective boards among the 2 inspected.

(a) all 10 selections: (1,2)(1,3)(1,4)(1,5)(2,3)(2,4)(2,5)(3,4)(3,5)(4,5)

(b) $P(X=0) = P\{(3,4)(3,5)(4,5)\} = 3/10 = .3$

$P(X=2) = P\{(1,2)\} = 1/10 = .1$

$P(X=1) = 1 - (.3 + .1) = .6$

(c) $F(0) = .30$

$F(1) = .90$

$F(2) = 1.00$

$F(x) = 0$	$x < 0$
.30	$0 \leq x < 1$
.90	$1 \leq x < 2$
1	$2 \leq x$

10. Worked: dice maximum M (p082-p083) - the elegant one

M = max of two dice

pmf: $p(1)=1/36, p(2)=3/36, p(3)=5/36, p(4)=7/36, p(5)=9/36, p(6)=11/36$

WHY the odd numbers? $M \leq m$ iff BOTH dice $\leq m \rightarrow m^2$ outcomes

$p(M = m) = [m^2 - (m-1)^2]/36 = (2m-1)/36$

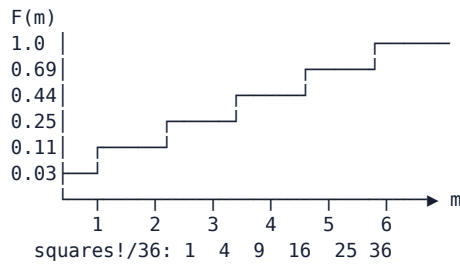
check $m=3$: $(2*3-1)/36 = 5/36 \checkmark$

cdf:

$F(m) = 0$	$m < 1$
1/36	$1 \leq m < 2$
4/36	$2 \leq m < 3$
9/36	$3 \leq m < 4$
16/36	$4 \leq m < 5$
25/36	$5 \leq m < 6$
1	$m \geq 6$

^ at integer support points $m=1, \dots, 6, F(m)=m^2/36$.

For real $1 \leq m < 6, F(m)=\text{floor}(m)^2/36$, as the steps above show.



11. Worked: cdf is non-decreasing (p085) - the proof question

SHOW: $x_1 < x_2 \rightarrow F(x_1) \leq F(x_2)$

proof:

$$\begin{aligned}
 F(x_2) &= P(X \leq x_2) \\
 &= P(\{X \leq x_1\} \cup \{x_1 < X \leq x_2\}) && \text{split the event} \\
 &= P(X \leq x_1) + P(x_1 < X \leq x_2) && \text{disjoint, so add} \\
 &\geq P(X \leq x_1) && \text{a probability is } \geq 0 \\
 &= F(x_1) && \checkmark
 \end{aligned}$$

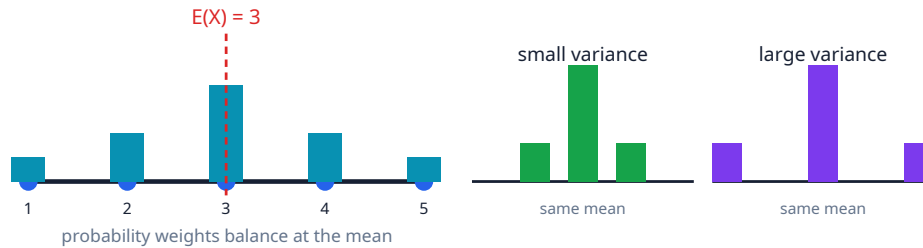
AND: $F(x_1) = F(x_2)$ exactly when $P(x_1 < X \leq x_2) = 0$
 (the interval between them carries no probability)

pmf and cdf are done. Next: expectation and variance (notes p086-p111).

04 EXPECTATION AND VARIANCE (notes p086-p111)

1. The three parameters (p086-p087)

Mean is the balance point; variance is squared spread



Expectation

$E[h(X)] = \text{sum or integral of } h(x) \text{ times weight}$

Variance

$\text{Var}(X) = E(X^2) - [E(X)]^2$

shifts preserve variance; multiplying by a scales variance by a^2

Mean as balance point and variance as spread

MEAN μ	centre / location
VARIANCE σ^2	spread
SD σ	spread, in the ORIGINAL units

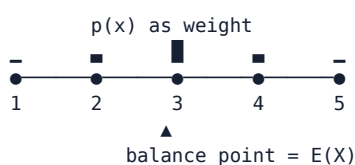
"knowledge of the numerical values of these parameters gives the researcher quick insight into the nature of the variable"

2. Expectation of a discrete rv (p088)

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

"the weighted average of the values, weighted by their probabilities"

the picture: a balance beam. values on the x-axis, probabilities as weights. the mean is the BALANCE POINT.



3. Worked: a fair die (p089)

$$p(x) = 1/6, \quad x = 1..6$$

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$$

$$= (1+2+3+4+5+6)/6 = 21/6 = 7/2 = 3.5$$

3.5 is not a face on the die. The mean need not be an achievable value. It is the long-run average.

4. Expectation of a function of X (p090)

$$E[h(X)] = \sum_{\{x\}} h(x) p(x) \quad (\text{discrete})$$

$$E[h(X)] = \int h(x) f(x) dx \quad (\text{continuous})$$

$$E[h(X)] \neq h(E(X)) \quad \text{in general}$$

you must apply h to EACH value, then weight, then sum.
you may NOT plug the mean into h.

5. Properties of the mean (p092-p093)

THEOREM $E[c] = c$ $E[cX] = c E[X]$

proof of $E[c]$:

$$E[c] = \sum c f(x) = c \sum f(x) = c(1) = c \quad \checkmark$$

proof of $E[cX]$:

$$E[cX] = \sum c x f(x) = c \sum x f(x) = c E[X] \quad \checkmark$$

EXERCISE (p093): prove $E[aX + b] = a E[X] + b$

$$E[aX+b] = \sum (ax+b)f(x) = a \sum x f(x) + b \sum f(x) \\ = a E[X] + b(1) = a E[X] + b \quad \checkmark$$

E is LINEAR, and it is linear with NO independence needed:
 $E[X+Y] = E[X] + E[Y]$ ALWAYS
 $E[aX+b] = aE[X]+b$ ALWAYS

6. Worked: the p094 triple (p094) - re-asked on ETE papers

$$\begin{array}{lcl} x & : & -3 \quad 6 \quad 9 \\ P(X=x) & : & 1/6 \quad 1/2 \quad 1/3 \end{array}$$

$$E(X) = (-3)(1/6) + 6(1/2) + 9(1/3) \\ = -0.5 + 3 + 3 = 11/2 = 5.5$$

$$E(X^2) = 9(1/6) + 36(1/2) + 81(1/3) \\ = 1.5 + 18 + 27 = 93/2 = 46.5$$

$$E((2X+1)^2) : \text{expand first} = E[4X^2 + 4X + 1] \\ = 4 E(X^2) + 4 E(X) + 1 \\ = 4(46.5) + 4(5.5) + 1 \\ = 186 + 22 + 1 = 209$$

THE METHOD: EXPAND, then use linearity term by term.
Do NOT compute the 3 values of $(2x+1)^2$ by hand first
(you can, but the algebra route is faster and safer).

sanity: $(2x+1)^2$ for $x=-3,6,9 = 25,169,361$
weighted: $25/6 + 169/2 + 361/3 = 4.17+84.5+120.3 = 209 \quad \checkmark$

7. Variance (p095-p100)

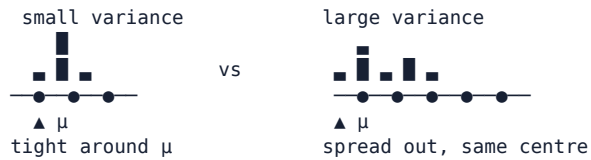
$$\begin{array}{lcl} \text{Var}(X) & = & \sigma^2 = E[(X - \mu)^2] \\ \text{SD} & = & \sigma = \sqrt{\text{Var}(X)} \end{array}$$

Var(X) is ALWAYS non-negative (it is an expectation of a SQUARE).

the idea: the SQUARED deviation from the mean, weighted by probability.
 large σ^2 $\Leftrightarrow$ more squared spread around μ
 small σ^2 $\Leftrightarrow$ less squared spread around μ

Variance measures spread. It is not a universal equivalence to predictability, and it is unrelated to the estimator property called consistency.

TWO DISTRIBUTIONS, SAME MEAN, DIFFERENT VARIANCE:



8. The shortcut formula (p101-p103) - and its proof

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

PROOF:

$$\begin{aligned} \text{Var}(X) &= E[(X-\mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] && \text{expand the square} \\ &= E[X^2] - 2\mu E[X] + \mu^2 && \text{linearity} \\ &= E[X^2] - 2\mu\mu + \mu^2 && \text{since } E[X] = \mu \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 && \checkmark \end{aligned}$$

THE TWO-STEP ROUTINE, ALWAYS:

1. $E(X)$
2. $E(X^2)$
3. $\text{Var} = E(X^2) - [E(X)]^2$

You almost never use the definition directly.

9. Variance transforms (p104-p106)

THEOREM $\text{Var}[c] = 0$ $\text{Var}[cX] = c^2 \text{Var}[X]$
 PROPOSITION $\text{Var}[aX+b] = a^2 \sigma^2$ $\sigma_{\{aX+b\}} = |a| \sigma_x$

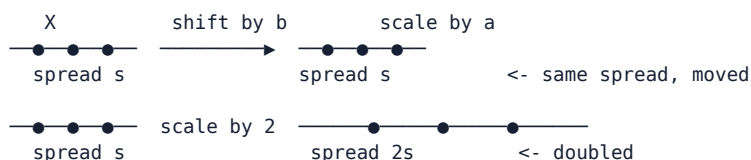
special cases:

$$\begin{aligned} \sigma_{\{aX\}} &= |a| \sigma_x && \text{scaling changes spread by } |a| \\ \sigma_{\{X+b\}} &= \sigma_x && \text{SHIFTING changes nothing} \end{aligned}$$

WHY $|a|$ AND NOT a

a may be negative, but a standard deviation cannot be.
 The sign of a does not affect spread, only scale.

the picture:



10. Worked: freezer dealer (p107-p108) - the 6-part chain

x	: 13.5	15.9	19.1	(cubic feet)
$p(x)$	: .2	.5	.3	

$$\begin{aligned} \text{(a) } E(X) &= 13.5(.2) + 15.9(.5) + 19.1(.3) \\ &= 2.70 + 7.95 + 5.73 = 16.38 \end{aligned}$$

$$\begin{aligned} E(X^2) &= 182.25(.2) + 252.81(.5) + 364.81(.3) \\ &= 36.45 + 126.405 + 109.443 = 272.298 \end{aligned}$$

$$V(X) = 272.298 - (16.38)^2 = 272.298 - 268.3044 = 3.9936$$

- (b) price = $25X - 8.5$
 $E(25X - 8.5) = 25(16.38) - 8.5 = 409.5 - 8.5 = 401$
- (c) $V(25X - 8.5) = (25)^2 V(X) = 625 \times 3.9936 = 2496$
 (the -8.5 vanishes from the variance)
- (d) actual capacity $h(X) = X - .01X^2$
 $E[h(X)] = E(X) - .01 E(X^2) = 16.38 - .01(272.298)$
 $= 16.38 - 2.723 = 13.657 \approx 13.66$

PART (d) IS THE TRAP PART.
 $E[X - .01X^2]$ is NOT $16.38 - .01(16.38)^2$.
 You must use $E(X^2)$, NOT $[E(X)]^2$.
 This is $E[h(X)] \neq h(E(X))$ in numerical form.

11. Worked: magazine orders (p109-p111) - the decision question

x	:	1	2	3	4	5	6
p(x)	:	1/15	2/15	3/15	4/15	3/15	2/15

store pays \$1 per copy, sells at \$2, leftovers worthless.
 is it better to order 3 or 4?

net revenue as a function of demand x:

x	:	1	2	3	4	5	6	
h3(x)	:	-1	1	3	3	3	3	<- order 3: plateaus at 3
h4(x)	:	-2	0	2	4	4	4	<- order 4: plateaus at 4
p(x)	:	1/15	2/15	3/15	4/15	3/15	2/15	

WHY h3 plateaus: order 3, cost 3, revenue 2 per copy SOLD (max 3 copies)

x=1: sold 1, revenue 2, cost 3 -> -1

x=2: sold 2, revenue 4, cost 3 -> +1

x=3: sold 3, revenue 6, cost 3 -> +3

x=4: can only sell 3, revenue 6, cost 3 -> +3 (plateau: demand > stock)

$$E[h3(X)] = (-1)(1/15) + (1)(2/15) + (3)(3/15) + (3)(4/15) + (3)(3/15) + (3)(2/15)$$

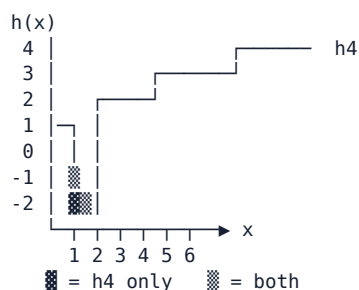
$$= (-1 + 2 + 9 + 12 + 9 + 6)/15 = 37/15 = 2.4667$$

$$E[h4(X)] = (-2)(1/15) + (0)(2/15) + (2)(3/15) + (4)(4/15) + (4)(3/15) + (4)(2/15)$$

$$= (-2 + 0 + 6 + 16 + 12 + 8)/15 = 40/15 = 2.6667$$

ORDER 4: higher expected revenue (2.67 vs 2.47)
 This is a DECISION question: express profit as h(X),
 take E of each option, compare.

the two profit curves:



Expectation and variance done. Next: continuous random variables (p112-p147).

05 CONTINUOUS RANDOM VARIABLES (notes p112-p147)

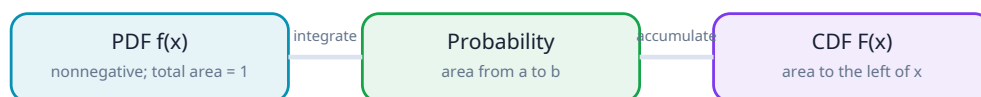
1. Why continuous needs new machinery (p113-p116)

- T = time of peak electricity demand at a power plant
- cannot be limited to a countable list of times
 - T can take ANY value in $[0, 24)$
 - $P(\text{peak at exactly } 12.013278...) = 0$ in a continuous model

A random variable is CONTINUOUS if it can assume any value in some interval, AND the probability of any specific single value is 0.

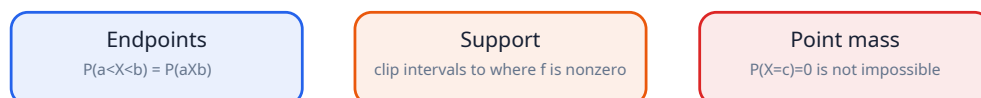
2. The probability density function, pdf (p117)

Continuous probability: area, accumulation, derivative



$$F(x) = \text{integral from } -\infty \text{ to } x \text{ of } f(t) \, dt$$

$f(x) = F'(x)$, where an absolutely continuous F is differentiable



continuous interval: $F(b) - F(a)$

integer-valued interval: $F(b) - F(a-1)$

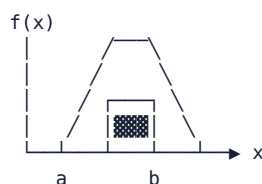
PDF, probability area, and CDF relationships

$f(x)$ is a pdf if:

- (i) $f(x) \geq 0$
- (ii) $\int_{-\infty}^{+\infty} f(x) \, dx = 1$
- (iii) $P(a \leq X \leq b) = \int_a^b f(x) \, dx$

$f(x)$ IS A DENSITY, NOT A PROBABILITY.
 $f(x)$ can be GREATER THAN 1.
 e.g. uniform on width 0.5 $\rightarrow f = 1/0.5 = 2$
 only its INTEGRAL is a probability.

$P(a \leq X \leq b) = \text{AREA under the density curve between } a \text{ and } b$:



the shaded AREA = $P(a \leq X \leq b)$

3. The five remarks (p119-p123) - each is a possible true/false

- R1 if $\int f = k$ and $k \neq 1$, then $f(x)/k$ IS the pdf
(this is how you "find k" - normalise)
- R2 X takes all values in (a,b) , where a,b may be $-\infty, +\infty$
- R3 $P[X = x_0] = \int \text{from } x_0 \text{ to } x_0 f \, dx = 0$ (zero width, zero area)
- R4 probability zero does NOT mean impossible for continuous rv
 $P(A) = 0$ does NOT imply $A = \emptyset$ <- subtle, examinable
- R5 if X only lives in $[a,b]$, set $f(x) = 0$ for all x outside $[a,b]$
(the SUPPORT convention)

R3 + R4 TOGETHER:
 $P(X = c) = 0$ for every point c
 yet SOME point must occur
 so "probability 0" and "impossible" are DIFFERENT here
 and $P(a < X < b) = P(a \leq X \leq b)$ exactly (endpoints free)

4. Worked: $f(x)=2x$ (p124-p125)

$f(x) = 2x$, $0 < x < 1$, 0 elsewhere

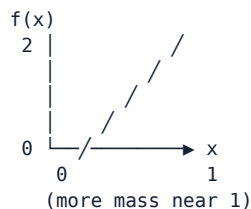
CHECK it is a pdf:

$$f(x) \geq 0 \quad \checkmark$$

$$\int_{-\infty}^{\infty} f = \int_0^1 2x \, dx = [x^2]_0^1 = 1 \quad \checkmark$$

$$P[X \leq 1/2] = \int_0^{1/2} 2x \, dx = [x^2]_0^{1/2} = 1/4$$

the density $f(x)=2x$ rises linearly:



5. Worked: continuous conditional (p126)

$$P[X \leq 1/2 \mid 1/3 \leq X \leq 2/3]$$

$$= \frac{P[1/3 \leq X \leq 1/2]}{P[1/3 \leq X \leq 2/3]} = \frac{\int_{1/3}^{1/2} 2x \, dx}{\int_{1/3}^{2/3} 2x \, dx}$$

$$= \frac{(1/4 - 1/9)}{(4/9 - 1/9)} = \frac{(5/36)}{(1/3)} = 5/12$$

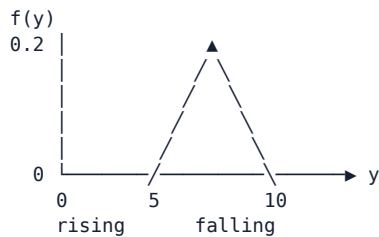
CONDITIONAL IN THE CONTINUOUS WORLD: fraction of AREAS.
 Same rule $P(A|B) = P(A \cap B) / P(B)$, but P 's are integrals.
 This exact shape is re-asked on the ETE S4 paper.

6. Worked: bus waiting time, Y (p127-p129) - the 6-part monster

two buses, each wait $\sim U(0,5)$, total wait Y has pdf:

$$f(y) = \begin{cases} y/25 & 0 \leq y < 5 \\ 2/5 - y/25 & 5 \leq y \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

the triangular density:



(b) VERIFY total area = 1:

$$\begin{aligned} & \int_0^5 y/25 \, dy + \int_5^{10} (2/5 - y/25) \, dy \\ &= [y^2/50]_0^5 + [2y/5 - y^2/50]_5^{10} \\ &= (25/50) + [(4 - 2) - (2 - 1/2)] \\ &= 1/2 + [2 - 1.5] = 1/2 + 1/2 = 1 \quad \checkmark \end{aligned}$$

$$(c) P(Y \leq 3) = \int_0^3 y/25 \, dy = [y^2/50]_0^3 = 9/50 = 0.18$$

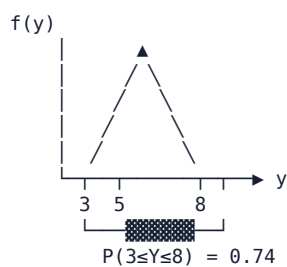
$$\begin{aligned} (d) P(Y \leq 8) &= \int_0^5 y/25 \, dy + \int_5^8 (2/5 - y/25) \, dy \\ &= 1/2 + [2y/5 - y^2/50]_5^8 \\ &= 1/2 + [(3.2 - 1.28) - (2 - 0.5)] \\ &= 1/2 + [1.92 - 1.5] = 1/2 + 0.42 = 0.92 \end{aligned}$$

$$(e) P(3 \leq Y \leq 8) = P(Y \leq 8) - P(Y < 3) = 46/50 - 9/50 = 37/50 = 0.74$$

$$\begin{aligned} (f) P(Y < 2 \text{ or } Y > 6) &= \int_0^2 y/25 \, dy + \int_6^{10} (2/5 - y/25) \, dy \\ &= 4/50 + 32/50 = 0.08 + 0.32 = 0.40 = 2/5 \end{aligned}$$

ERRATA 13: the slide's working for (f) prints the first integral bound as 3, not 2, which contradicts its own stated event " $Y < 2$ ". The final answer $2/5 = 0.40$ is still correct for " $Y < 2$ or $Y > 6$ ". Use $0.08 + 0.32 = 0.40$.

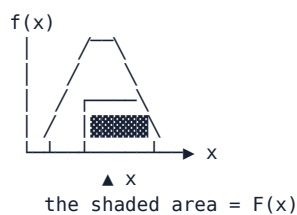
the (e) picture:



7. The cdf for continuous (p131-p134)

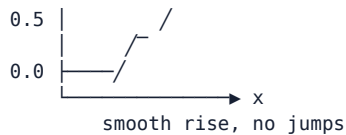
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) \, dy$$

$F(x)$ = the AREA to the LEFT of x



$F(x)$ is a smooth S-curve (not a staircase like the discrete cdf):





8. Using F(x) (p134-p135)

$$P(X > a) = 1 - F(a)$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

CONTINUOUS vs INTEGER: the SAME theorem has two forms

continuous : $P(a \leq X \leq b) = F(b) - F(a)$

integer rv : $P(a \leq X \leq b) = F(b) - F(a-1)$ <- the a-1

why the a-1:

continuous: $F(a)$ excludes the zero-width point a -> fine

integer : $F(a)$ INCLUDES the value a -> must drop it

9. Getting f from F (p136-p137)

for discrete : pmf = DIFFERENCE of two cdf values

for an absolutely continuous distribution: pdf = DERIVATIVE of the cdf

$$F'(x) = f(x) \quad \text{where } F \text{ is differentiable}$$

10. Worked: find the cdf, exercise 1 (p138)

$$(i) \quad f(x) = \begin{cases} 1/3 & 0 \leq x \leq 1 \\ 2/3 & 1 < x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{CHECK: } \int_{-\infty}^{\infty} f(x) dx = \int_0^1 1/3 dx + \int_1^2 2/3 dx = 1/3 + 2/3 = 1 \quad \checkmark$$

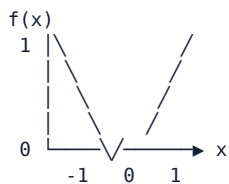
$$F(x) = \begin{cases} 0 & x < 0 \\ x/3 & 0 \leq x \leq 1 & (\int_{-\infty}^x 1/3 du = x/3) \\ 1/3 + 2(x-1)/3 & 1 < x \leq 2 & (1/3 + \int_1^x 2/3 du) \\ 1 & x > 2 \end{cases}$$

$$(ii) \quad f(x) = |x|, \quad -1 < x < 1$$

split at 0 because of the absolute value:

$$F(x) = \begin{cases} 0 & x \leq -1 \\ \int_{-1}^x (-u) du & -1 < x \leq 0 & = (1 - x^2)/2 \\ \int_{-1}^0 (-u) du + \int_0^x u du & 0 < x < 1 & = (1 + x^2)/2 \\ 1 & x \geq 1 \end{cases}$$

the $|x|$ density is a V:



11. Worked: cdf to pdf (p139) - the reverse direction

$$F(x) = \begin{cases} 0 & x < 1 \\ (x-1)^4 / 16 & 1 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

(i) DENSITY = derivative:

$$f(x) = d/dx [(x-1)^4 / 16] = 4(x-1)^3 / 16 = (x-1)^3 / 4, \quad 1 \leq x \leq 3$$

0 elsewhere

$$(ii) \quad P(2 \leq X \leq 3) = F(3) - F(2)$$

$$= 1 - (2-1)^{4/16}$$

$$= 1 - 1/16 = 15/16$$

USING F IS THE FAST ROUTE. If the question gives F, do NOT integrate the density you just found. Just subtract two F values.

12. Expected values, continuous (p140-p144)

summation $\rightarrow$ integration
pmf $\rightarrow$ pdf

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

$$E(aX+b) = a E(X) + b \quad (\text{same linear rule})$$

$$\text{Var}(X) = \int (x-\mu)^2 f(x) dx = E(X^2) - [E(X)]^2$$

$$\sigma = \sqrt{\text{Var}(X)}$$

13. Worked: hospitalization, $Y=X+4$ (p142) - and its ETE sibling

$Y = X + 4$, where $f(x) = 32/(x+4)^3$, $x > 0$
find $E(Y)$

$$E(Y) = E(X+4) = E(X) + 4$$

$$E(X) = \int_0^{\infty} x \cdot 32/(x+4)^3 dx$$

substitute $u = x+4$, $x = u-4$, $dx = du$, limits $4 \rightarrow \infty$:

$$= 32 \int_4^{\infty} (u-4)/u^3 du$$

$$= 32 \int_4^{\infty} (u^{-2} - 4u^{-3}) du$$

$$= 32 [-u^{-1} + 2u^{-2}]_4^{\infty}$$

$$= 32 [0 - (-1/4 + 2/16)]$$

$$= 32 [1/4 - 1/8] = 32 (1/8) = 4$$

$$E(Y) = 4 + 4 = 8 \text{ days}$$

THE TRICK: $Y = X + 4$, so use the SHIFT rule.
Do NOT integrate $(x+4)$ against the density.
 $E(X+4) = E(X) + 4$ is one line.

14. Worked: Pareto density (p146-p147)

$$f(x) = k/x^{(k+1)}, \quad x \geq 1, \quad k > 0$$

$$\text{normalise: } \int_1^{\infty} k x^{-(k+1)} dx = k [-x^{-k}/k]_1^{\infty} = 1 \quad \checkmark$$

$E(X)$ exists only if $k > 1$ (otherwise the integral diverges)
 Var exists only if $k > 2$

THE CONVERGENCE CONDITION IS EXAMINABLE.
A heavy tail can make $E(X)$ or $\text{Var}(X)$ INFINITE.
Always check the exponent condition before claiming a mean.

15. The continuous toolkit, one block

check a pdf	$\int \text{over support} = 1$
find k	set up that integral = 1, solve
$P(a \leq X \leq b)$	$\int_a^b f$, or $F(b) - F(a)$
cdf	$\int_{-\infty}^x f$
pdf from cdf	$f = F'$

E(X)	$\int x f(x) dx$	
Var(X)	$E(X^2) - [E(X)]^2$	
conditional	ratio of two integrals	

That is the whole continuous part. Next: the two named discrete distributions.

06 BINOMIAL AND POISSON (ppt3, 28 pages, lectures 12-13)

PART A: BINOMIAL

1. The derivation from 5 coin tosses (p003-p007)

QUESTION: P(exactly 3 heads in 5 tosses)?

one arrangement: HHHTT

$$P = (1/2)(1/2)(1/2)(1/2)(1/2) = (1/2)^3 (1/2)^2$$

another: THHHT

$$P = (1/2)(1/2)(1/2)(1/2)(1/2) = (1/2)^3 (1/2)^2$$

EVERY arrangement with 3 H and 2 T has the SAME probability.

how many arrangements? 5 positions, choose 3 for H

$$C(5,3) = 5!/(3! 2!) = 120/(6 \times 2) = 10$$

the 10 arrangements, all equally likely:

THHHT	HHHTT	TTHHH	HTTHH	HHTTH
HTHHT	HTHTH	HTHTH	HHHTT	THHHT

THEREFORE:

$$\begin{aligned} P(3 \text{ heads and 2 tails}) &= C(5,3) \times P(H)^3 \times P(T)^2 \\ &= 10 \times (1/2)^5 \\ &= 10/32 = 0.3125 = 31.25\% \end{aligned}$$

THE STRUCTURE, IN ONE PICTURE:

[number of arrangements]	x	[probability of one]
$C(n,x)$		$p^x q^{(n-x)}$

2. The setup conditions (p008-p011)

a random experiment performed repeatedly

"success" = the event occurs, "failure" = it does not

PHYSICAL CONDITIONS FOR A BINOMIAL

1. each trial gives TWO mutually disjoint outcomes
2. n, the number of trials, is FINITE
3. the trials are INDEPENDENT
4. p, the success probability, is CONSTANT

$$q = 1 - p \quad (\text{failure probability})$$

3. The definition (p009-p010)

$$\begin{aligned} P(X = x) &= C(n,x) p^x q^{(n-x)}, & x = 0,1,2,\dots,n \\ P(X = x) &= 0 & \text{otherwise} \end{aligned}$$

n and p are the PARAMETERS. n is also called the DEGREE.

NOTATION: $X \sim B(n, p)$

4. Mean and variance

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= npq \\ \text{SD} &= \sqrt{npq} \end{aligned}$$

5. WORKED: pens (p013-p016) - the flagship example

10% of pens are defective. A box has 12 pens.

X = number of defective pens in the box.
 $n = 12, p = 0.1, q = 0.9 \quad X \sim B(12, 0.1)$

(i) EXACTLY 2:

$$\begin{aligned} P(X=2) &= C(12,2) (0.1)^2 (0.9)^{10} \\ &= 66 \times 0.01 \times 0.3487 \\ &= 0.2301 \end{aligned}$$

(ii) AT LEAST 2 (use the COMPLEMENT):

$$P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$$

$$P(X=0) = C(12,0) (0.1)^0 (0.9)^{12} = 0.2824$$

$$P(X=1) = C(12,1) (0.1)^1 (0.9)^{11} = 12 \times 0.1 \times 0.3138 = 0.3766$$

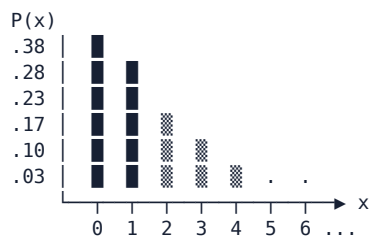
$$P(X \geq 2) = 1 - [0.2824 + 0.3766] = 1 - 0.659 = 0.341$$

(iii) NONE:

$$P(X=0) = 0.2824$$

THE "AT LEAST 2" MOVE IS THE WHOLE POINT OF THIS EXAMPLE.
 Summing $P(2)+P(3)+\dots+P(12)$ is 11 terms.
 Complementing is 2 terms. ALWAYS complement for "at least".

the pmf picture for $B(12, 0.1)$:



$P_0 = 0.28, P_1 = 0.38$ <- note $P_1 > P_0$ here ($p=0.1, n=12$, mode at $x=1$)

6. WORKED: irregular die (p017-p018) - the parameter recovery

A six-faced die. $P(5 \text{ evens in } 10 \text{ throws}) = 2 \times P(4 \text{ evens in } 10 \text{ throws})$.
 In 10,000 sets of 10 throws, expected number with NO even number?

set the condition:

$$C(10,5) p^5 q^5 = 2 \times C(10,4) p^4 q^6$$

$$252 p^5 q^5 = 2 \times 210 p^4 q^6$$

$$252 p = 420 q \quad (\text{divide by } p^4 q^5)$$

$$3p = 5q = 5(1-p)$$

$$3p = 5 - 5p$$

$$8p = 5$$

$$p = 5/8, \quad q = 3/8$$

then:

$$P(X=0) = C(10,0) (5/8)^0 (3/8)^{10} = (3/8)^{10} = 0.00005499$$

$$\begin{aligned} \text{expected number out of } 10,000 \\ &= 10000 \times 0.00005499 = 0.5499 \end{aligned}$$

TWO SKILLS IN ONE QUESTION:
 1. recover p from a ratio $P(X=k+1) = c P(X=k)$
 (the C's and the powers of q cancel, leaving a linear eq)
 2. EXPECTED COUNT = $N \times P$ (not P alone!)
 0.5499, NOT 0.00005499

ERRATA 2: the slide prints 0.549; the exact value is 0.5499.

PART B: POISSON

7. What it models (p019-p021)

Poisson models the number of "RARE EVENTS" occurring randomly in a fixed interval.

the slide's seven instances:

1. deaths from a disease (heart attack, cancer, snake bite)
2. suicides reported in a city
3. defective material in a packing
4. faulty blades in a packet of 100
5. air accidents in a unit of time
6. printing mistakes per page of a book
7. cars passing a crossing per minute (busy hours)

8. Poisson as the limit of binomial (p022)

under these three conditions:

1. $n \rightarrow \infty$ indefinitely large
2. $p \rightarrow 0$ indefinitely small
3. $np = \lambda$, finite $p = \lambda/n$, $q = 1 - \lambda/n$

$$C(n, x) p^x q^{n-x} \xrightarrow{\text{limit}} e^{-\lambda} \lambda^x / x! \quad \text{with } \lambda = np$$

"USE THE APPROXIMATION" IN A QUESTION MEANS:
Poisson with $\lambda = np$, NOT the exact binomial.
e.g. 5000 men, $p=0.001 \rightarrow \lambda = 5$

9. The definition (p023-p025)

$$P(X = x) = e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots, \lambda > 0$$

$$P(X = x) = 0 \quad \text{otherwise}$$

λ is the PARAMETER. NOTATION: $X \sim P(\lambda)$

$$E(X) = \lambda \quad \text{Var}(X) = \lambda \quad \leftarrow \text{MEAN EQUALS VARIANCE}$$

A POISSON VARIABLE HAS MEAN = VARIANCE
Equality is necessary for Poisson, but it does not prove that
an otherwise unspecified distribution is Poisson. Use the
model assumptions or a stated Poisson family as well.

10. WORKED: parameter from a ratio (p025)

$$P(X=1) = 0.2 P(X=2). \quad \text{Find } P(X=0).$$

$$\frac{e^{-\lambda} \lambda^1}{1!} = 0.2 \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\lambda = 0.2 \lambda^2 / 2 = 0.1 \lambda^2$$

$$1 = 0.1 \lambda \quad (\text{divide by } \lambda, \lambda > 0)$$

$$\lambda = 10$$

then:

$$P(X=0) = e^{-10} 10^0 / 0! = e^{-10} = 0.0000454$$

the cancellation picture:

$$\begin{array}{ccc} \underbrace{e^{-\lambda} \lambda}_\text{cancels} & = & 0.2 \times \underbrace{e^{-\lambda} \lambda^2}_\text{cancels} \\ \lambda & = & 0.2 \lambda^2 / 2 \\ \longrightarrow & & \lambda = 10 \end{array}$$

11. WORKED: telephone calls, THE NESTING (p026)

calls arrive at $\lambda = 2$ per minute.

$P(\text{exactly two calls in each of the first 5 minutes})?$

STEP 1: Poisson within ONE minute

$$P(N=2) = e^{-2} 2^2 / 2! = 2e^{-2} = 0.270671$$

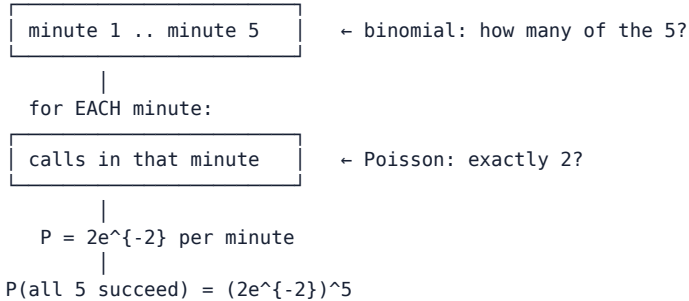
STEP 2: binomial over the FIVE minutes

M = number of minutes (of 5) with exactly 2 calls

$M \sim B(5, 0.270671)$

$$P(M=5) = (2e^{-2})^5 = 32 e^{-10} = 0.00145$$

THE STRUCTURE:



THIS IS THE COMPOSITION (SLOT 9) QUESTION SHAPE.
 The inner model gives a probability. That probability then becomes the p of an outer binomial.
 Recognise: "in EACH of the k intervals" -> nest.

12. WORKED: life insurance (p027)

5000 men aged 42, $P(\text{death in a year}) = 0.001$.

$P(\text{exactly 4 claims})?$

$$\lambda = n p = 5000 \times 0.001 = 5$$

$$P(X=4) = e^{-5} \frac{5^4}{4!} = e^{-5} \frac{625}{24} \\ = 0.0067379 \times 26.0417 = 0.175467$$

ERRATA 1: the slide prints 0.1745. The correct value rounds to 0.1755.

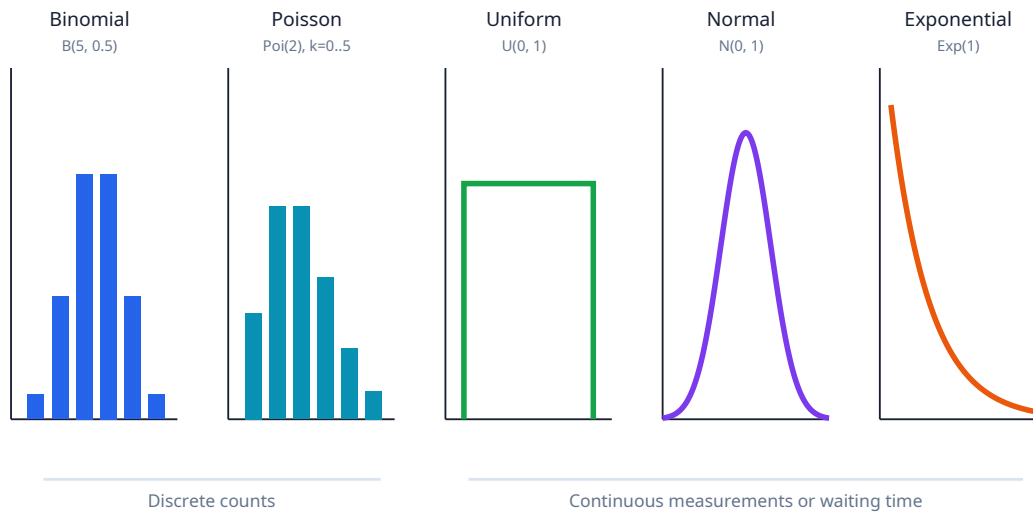
13. The tail form (p028)

$$P(X > 2) = 1 - P(0) - P(1) - P(2)$$

$$\text{general: } P(X > k) = 1 - \sum_{i=0}^k P(X=i)$$

PART C: THE DISCRETE DISTRIBUTIONS SIDE BY SIDE

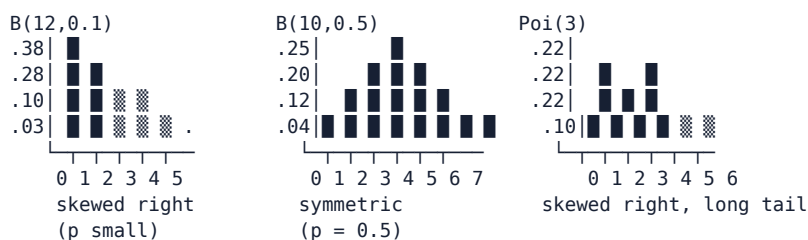
Five distribution shapes



Distribution shapes, with the discrete models on the left

	BINOMIAL	POISSON
models	# successes in n trials	# rare events in a fixed interval
setup	n finite, indep, p constant	rare, indep in time/space
pmf	$C(n, x)p^x q^{n-x}$	$e^{-\lambda} \lambda^x / x!$
mean	np	λ
variance	npq	λ
signature	npq < np always	mean = variance
link	$n \rightarrow \infty, p \rightarrow 0, np = \lambda \rightarrow$	(the limit)

the pmf shapes:



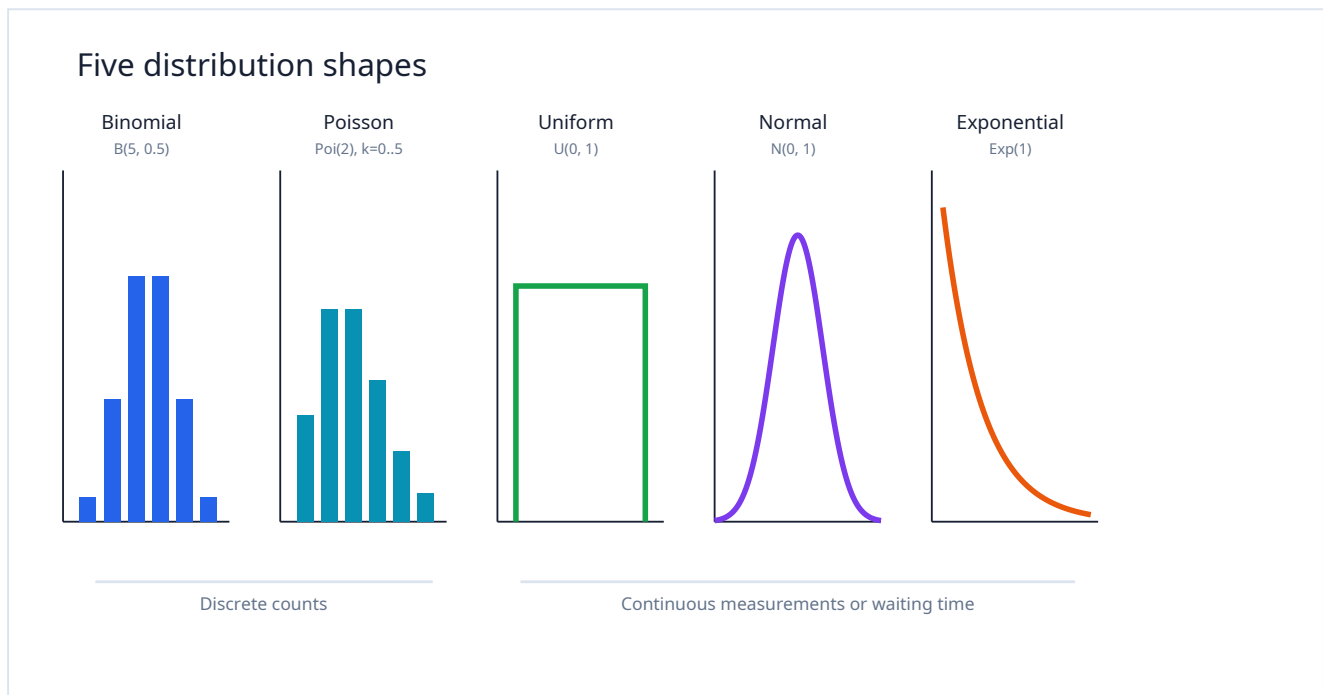
THE RECOGNITION ROUTINE FOR THESE TWO

"n trials, two outcomes, independent, p constant"
 "arrivals, calls, defects, rare, per unit time"
 "use the approximation"
 "in each of the k intervals"
 "mean equals variance"
 "at least one"

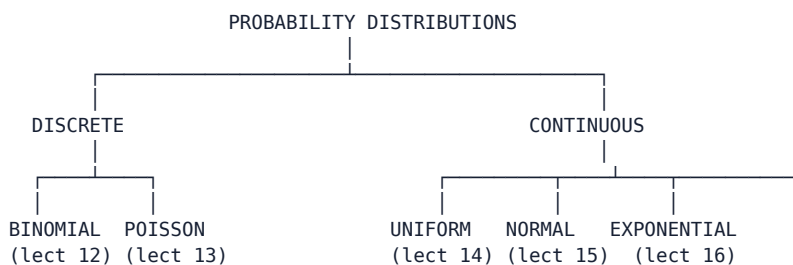
-> BINOMIAL
 -> POISSON
 -> POISSON with $\lambda = np$
 -> NEST (Poisson then binomial)
 -> necessary Poisson check;
 confirm the model too
 -> complement, $1 - q^n$

07 UNIFORM, NORMAL, EXPONENTIAL (ppt4, 44 pages, lectures 14-16)

The family tree first (p001)



Five distribution shapes for comparison



PART A: UNIFORM (p002-p006) - lecture 14

1. Definition

constant density, so equal-length subintervals have equal probabilities. Individual points each have probability zero.

$$\begin{aligned} f(x) &= 1/(b-a) & a \leq x \leq b \\ f(x) &= 0 & \text{otherwise} \end{aligned}$$

the flat rectangle:



the area is width $\times$ height = $(b-a) \times 1/(b-a) = 1$ ✓

2. Properties

$$\mu = (a + b)/2$$

$$\sigma^2 = (b - a)^2/12 \quad (\text{the 12 is FIXED})$$

THE LENGTH-RATIO SHORTCUT: probability = overlap length divided by total length. No integration needed.
 $P(c \leq X \leq d) = \text{length}([c,d] \text{ intersection } [a,b]) / (b-a)$
 If $[c,d]$ lies inside $[a,b]$, this becomes $(d-c)/(b-a)$.

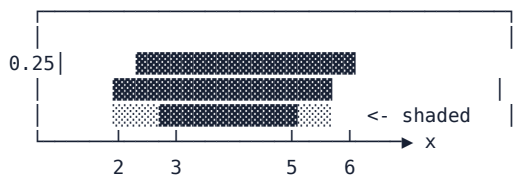
3. WORKED: U(2,6) (p004-p006)

$$f(x) = 1/(6-2) = 0.25, \quad 2 \leq x \leq 6$$

$$\mu = (2+6)/2 = 4$$

$$\sigma^2 = (6-2)^2/12 = 16/12 = 1.333$$

$$P(3 < X < 5) = \text{length ratio} = (5-3)/(6-2) = 2/4 = 0.50$$



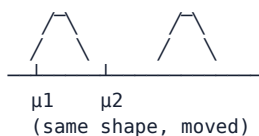
$$P = \text{shaded} / \text{total} = 2/4 = 0.50$$

PART B: NORMAL (p007-p037) - lecture 15

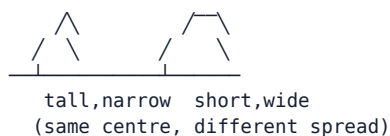
4. Shape facts (p008-p010)

BELL SHAPED
 SYMMETRICAL
 MEAN = MEDIAN = MODE = μ
 range: $-\infty$ to $+\infty$
 location set by μ , spread set by σ

changing μ shifts:



changing σ widens:



5. The notation and the density (p010-p011)

$$X \sim N(\mu, \sigma^2)$$

THE SECOND SLOT IS THE ****VARIANCE****, not the SD
 "mean 70, variance 25" $\rightarrow \mu=70, \sigma^2=25, \sigma=5$
 ALWAYS square-root it before a z-score.

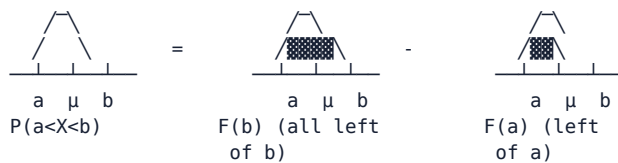
$$f(x) = 1/(\sigma\sqrt{2\pi}) \cdot e^{-\{(x-\mu)^2/(2\sigma^2)\}}$$

6. The cdf and interval (p012-p013)

$$F(x_0) = P(X \leq x_0) \quad (\text{area to the left})$$

$$P(a < X < b) = F(b) - F(a)$$

the subtraction picture:

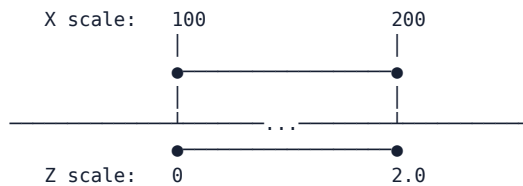


7. Standardisation (p014-p015)

$$Z = (X - \mu) / \sigma \quad Z \sim N(0, 1)$$

WORKED: $\mu=100$, $\sigma=50$, $X=200$
 $Z = (200-100)/50 = 2.0$

the two scales, same shape:



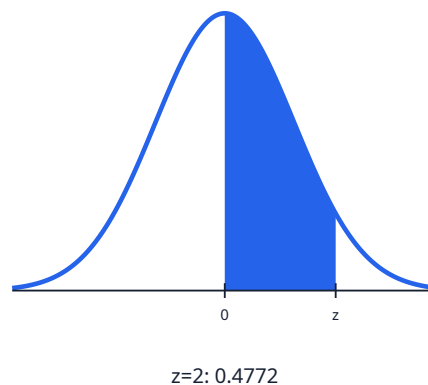
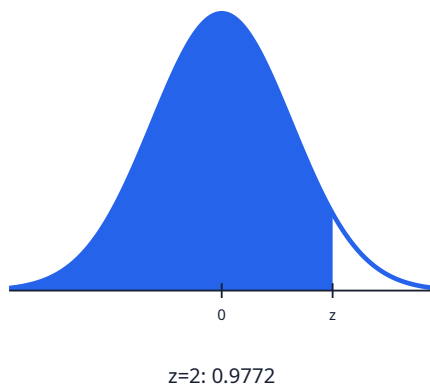
"the distribution is the same, only the SCALE has changed"

8. THE TWO TABLES - the biggest trap in the whole paper (p022-p023)

Two normal-table conventions

p022: cumulative $P(Z < z)$

p023: centre area $P(0 < Z < z)$



Cumulative and centre-area normal tables

THE SLIDES SHOW **TWO DIFFERENT TABLES** AND THEY LOOK ALIKE

TABLE 1 (p022, cumulative standard normal table):
 gives $P(Z < z)$ = CUMULATIVE from $-\infty$ to z
 $F(z)$. e.g. $z=0.0 \rightarrow 0.5000$, $z=2.0 \rightarrow 0.9772$

TABLE 2 (p023, area from 0 to z):
 gives $P(0 < Z < z)$ = area FROM THE CENTRE to z
 $A(z)$. e.g. $z=2.0 \rightarrow 0.4772$

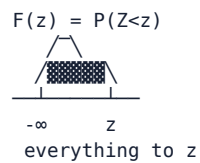
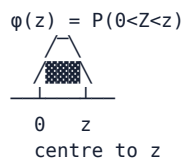
THEY DIFFER BY 0.5 FOR $z > 0$:

$$F(z) = 0.5 + A(z)$$

$$0.9772 = 0.5 + 0.4772 \quad \checkmark$$

DECIDE WHICH TABLE THE QUESTION GIVES **BEFORE** SUBSTITUTING

the two areas, drawn:



9. Symmetry facts (p024-p025)

$$P(Z < -2.00) = 1 - P(Z < 2.00) = 1 - 0.9772 = 0.0228$$

$$\Phi(-z) = 1 - \Phi(z)$$

$$P(Z > z) = 1 - \Phi(z)$$

the mirror picture:



the left tail at $-z$ EQUALS the right tail at $+z$

10. WORKED: the 8.6 chain (p028-p033)

$$X \sim N(8.0, 5.0^2) \quad [\text{note: } \sigma = 5.0]$$

(a) FIND $P(X < 8.6)$:

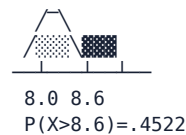
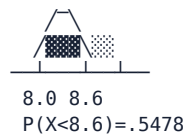
$$Z = (8.6 - 8.0)/5.0 = 0.12$$

$$P(X < 8.6) = P(Z < 0.12) = F(0.12) = 0.5478$$

(b) FIND $P(X > 8.6)$:

$$P(X > 8.6) = 1 - 0.5478 = 0.4522$$

the two areas:



ERRATA 8: the figure on p030 labels the curve " $\sigma = 10$ ", but the arithmetic on the same page divides by 5.0. Use $\sigma = 5.0$.

11. WORKED: the inverse, 20% lower tail (p034-p037)

Find x so that only 20% of values are BELOW x .

STEP 1: find z for a 20% lower tail

$$\text{look for } F(z) = 0.20$$

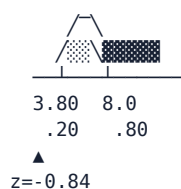
$$\text{from the table, } z \approx -0.84 \quad (\text{exactly } -0.8416)$$

STEP 2: CONVERT BACK TO X ← the forgotten step

$$X = \mu + Z\sigma = 8.0 + (-0.84)(5.0) = 8.0 - 4.2 = 3.80$$

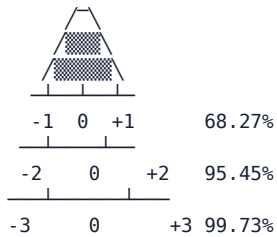
THE INVERSE HAS TWO STEPS AND PEOPLE STOP AFTER STEP 1.
Getting $z = -0.84$ is HALF the answer.
The mark is in $X = \mu + z\sigma$.

the picture:



12. The landmarks (the hidden item H4)

within 1σ : 68.27%
 within 2σ : 95.45%
 within 3σ : 99.73%



PART C: EXPONENTIAL (p038-p043) - lecture 16

13. What it models

the LENGTH OF TIME between two occurrences of an event
 (the time between arrivals)

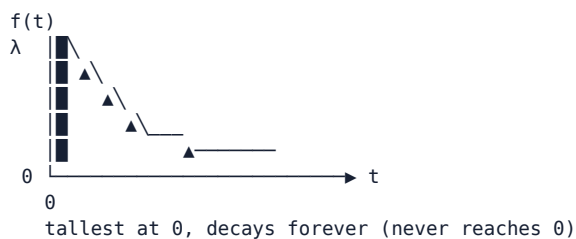
examples: trucks at a dock, transactions at an ATM, phone calls

14. The density and its parameters (p040-p041)

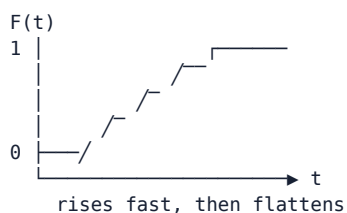
$$\begin{aligned}
 f(t) &= \lambda e^{-\lambda t}, & t > 0 \\
 \mu &= 1/\lambda & \sigma^2 &= 1/\lambda^2 \\
 F(t) &= 0 & \text{for } t \leq 0 \\
 &= 1 - e^{-\lambda t} & \text{for } t > 0 \\
 P(T > t) &= 1 & \text{for } t < 0 \\
 &= e^{-\lambda t} & \text{for } t \geq 0
 \end{aligned}$$

ERRATA 4: the slide on p040 says "1/λ is the MEAN NUMBER of occurrences per unit time". THAT IS BACKWARDS.
 λ = the rate (occurrences per unit time)
 $1/\lambda$ = the mean TIME between occurrences
 p041 states it correctly as $\mu = 1/\lambda$.

THE DECAY PICTURE:



the cdf rises toward 1:



15. WORKED: arrivals at 15 per hour (p042-p043)

"customers arrive at the rate of 15 per hour.
 $P(\text{the arrival time between consecutive customers} < 3 \text{ minutes})$?"

$\lambda = 15$ per HOUR

```

3 minutes = 0.05 hours      <- CONVERT THE UNIT
P(T < 0.05) = 1 - e^{-λt}
              = 1 - e^{-(15)(0.05)}
              = 1 - e^{-0.75}
              = 1 - 0.4724 = 0.5276

```

THE UNIT CONVERSION IS THE WHOLE QUESTION.
 λ is per hour, so t MUST be in hours.
 3 min = 0.05 hr.
 This is the trap the setter builds into almost every exponential story (errata families E24S4, A2 D2).

the mean gap = $1/\lambda = 1/15$ hour = 4 minutes

16. The memoryless property (hidden item H3)

$$P(T > s + t \mid T > s) = P(T > t)$$

"surviving s MORE, given you already survived to now, is the same as surviving s from scratch"
 the exponential NEVER AGES. A used part is as good as new. (unlike, say, a lightbulb that wears out)

PROOF SKETCH:

$$\begin{aligned}
 P(T > s+t \mid T > s) &= P(T > s+t \text{ and } T > s) / P(T > s) \\
 &= P(T > s+t) / P(T > s) \\
 &= e^{-\lambda(s+t)} / e^{-\lambda s} \\
 &= e^{-\lambda t} = P(T > t) \quad \checkmark
 \end{aligned}$$

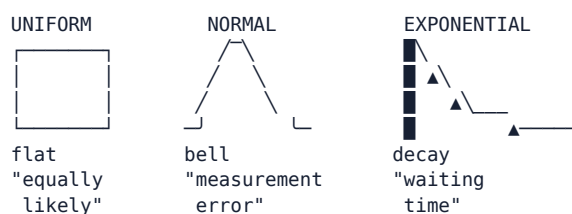
the $e^{-\lambda s}$ cancels <- the whole trick

the canonical number: $e^{-1} = 0.3679$

THE THREE CONTINUOUS DISTRIBUTIONS, ONE TABLE

	UNIFORM	NORMAL	EXPONENTIAL
pdf	$1/(b-a)$	$1/(\sigma\sqrt{2\pi})e^{..}$	$\lambda e^{-\lambda t}$
mean	$(a+b)/2$	μ	$1/\lambda$
variance	$(b-a)^2/12$	σ^2	$1/\lambda^2$
cdf	0, ratio, 1	table	0 or $1-e^{-\lambda t}$
tail	ratio	$1-\Phi(z)$	$e^{-\lambda t}$
parameter	a, b	μ, σ^2	λ (a RATE)
models	random pt	measurement	waiting time

the three shapes side by side:



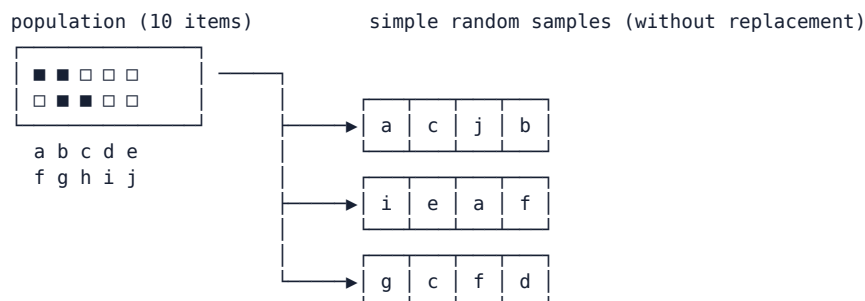
08 SAMPLING AND THE CENTRAL LIMIT THEOREM (lms-standard-error-clt, 19 pages, lectures 17-18)

1. Population vs sample (p001-p002)

POPULATION (N)	SAMPLE (n)
the ENTIRE group you want to draw conclusions about	a smaller subset, collected to observe and analyse
described by PARAMETERS μ , σ	described by STATISTICS $\bar{X}$, s
often too large / costly / impossible to measure	must be representative and randomly selected
e.g. all 500,000 bulbs	e.g. 100 randomly chosen

PARAMETER vs STATISTIC - the words the exam checks
 PARAMETER = a number about the POPULATION (μ , σ , p)
 STATISTIC = a number computed FROM THE SAMPLE ($\bar{X}$, s , $p^{\wedge}$)
 The parameter is UNKNOWN (that is why we sample).
 The statistic is KNOWN (we compute it).

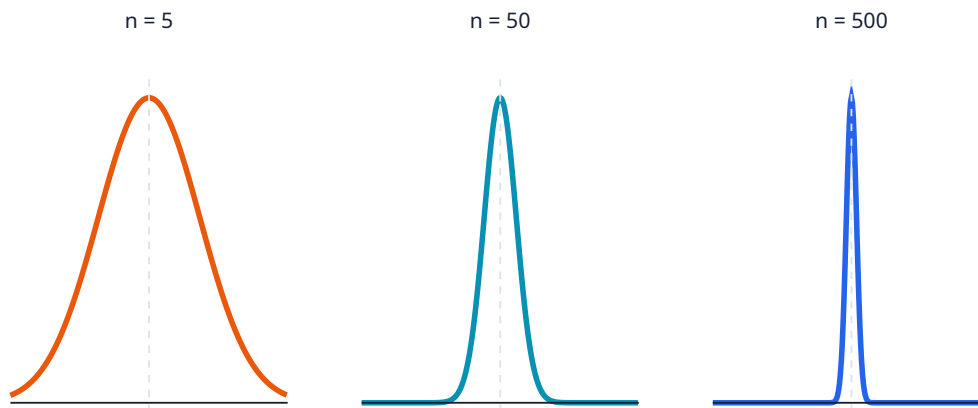
2. Simple random sampling, the picture (p003)



■ = the sampled items are spread randomly, NOT clustered
 "without replacement" = once picked, an item is not put back

3. Standard error (p004)

Sampling distributions narrow as n grows



centre stays at μ ; widths use the exact $1 / \sqrt{n}$ ratio

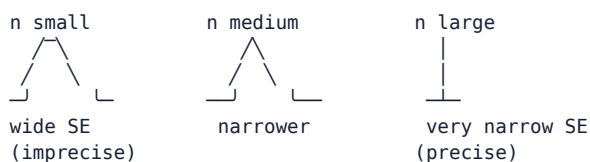
Sampling distributions narrow at the exact square-root rate

STANDARD ERROR measures the variability of a STATISTIC (usually the sample mean) across multiple hypothetical samples. It quantifies how much the estimate deviates from the true parameter.

$$SE = \sigma / \sqrt{n}$$

KEY: as n increases, SE DECREASES.
Larger samples $\rightarrow$ more precise estimates.

the shrinking spread:



4. WORKED: lightbulbs (p005)

$\sigma = 100$ hours.

Case 1: $n = 25$ $SE = 100/\sqrt{25} = 100/5 = 20$ hours

Case 2: $n = 100$ $SE = 100/\sqrt{100} = 100/10 = 10$ hours

THE SQRT RULE, memorise it:
QUADRUPLING n HALVES the SE
25 $\rightarrow$ 100 is $\times 4$, so SE goes 20 $\rightarrow$ 10 (halved)
this is the answer to every "how large a sample" question

5. The Central Limit Theorem (p007-p010)

"If you take sufficiently large samples from a population, the distribution of the SAMPLE MEANS will follow a normal distribution, REGARDLESS of the population's underlying

shape (even if heavily skewed, uniform, or bimodal)."

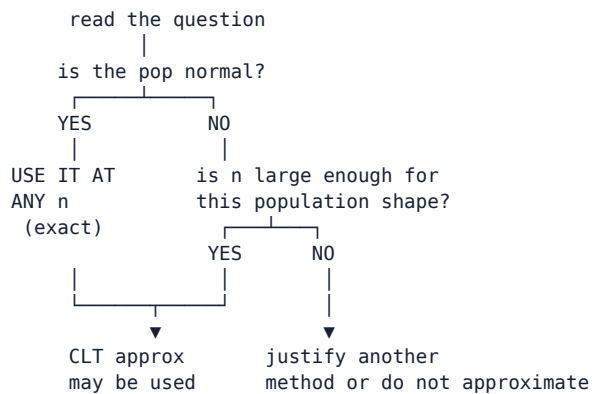
THREE RULES:

1. CENTRE mean of the sample means = μ ($\mu_{\bar{X}} = \mu$)
2. SPREAD sd of the sample means = $\sigma/\sqrt{n}$ ($\sigma_{\bar{X}} = \sigma/\sqrt{n}$)
3. SHAPE as n increases the shape becomes normal;
 $n \geq 30$ is the course rule of thumb

6. THE THREE CASES - the applicability conditions (p008)

1. NON-NORMAL population, large n
-> sample means may be approximately Normal(μ , $\sigma/\sqrt{n}$)
 $n \geq 30$ is the course heuristic, not a theorem cutoff
2. $n \leq 30$ AND the population IS normal
-> sample means are EXACTLY Normal(μ , $\sigma/\sqrt{n}$)
3. Small n AND the population is NOT normal
-> do not assume a normal approximation without evidence

the decision tree:



7. The visual proof (p010a-p010b)

the 3x4 grid the slide shows:

	NORMAL	UNIFORM	SKEWED	RANDOM
POPULN				
$n = 10$				
$n = 30$				

^ all four columns look NARROW BELLS at $n=30$
that is the CLT: shape washed out, spread shrunk

8. WORKED: ATM wait times (p011-p012)

heavily right-skewed wait, $\mu = 4$ min, $\sigma = 2$ min, $n = 36$.
 $P(\text{average wait} > 4.5 \text{ min})?$

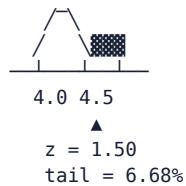
STEP 1 $n=36 \geq 30$, so the sample mean is normal (even though waits are skewed)

STEP 2 $SE = 2/\sqrt{36} = 2/6 = 0.333$

STEP 3 $Z = (\bar{X} - \mu)/SE = (4.5 - 4)/0.333 = 0.5/0.333 = 1.50$

STEP 4 upper tail at $z=1.50$ -> 6.68%

the picture:



9. WORKED: impurity (p013-p014) - the ERRATA one

$\mu = 4.0 \text{ g}$, $\sigma = 1.5 \text{ g}$, $n = 50$.
 $P(3.5 < \bar{X} < 3.8)$?

$$SE = 1.5/\sqrt{50} = 0.212132$$

$$Z1 = (3.5 - 4)/0.212132 = -2.357 \rightarrow -2.36$$

$$Z2 = (3.8 - 4)/0.212132 = -0.943 \rightarrow -0.94$$

$$P = P(0 < Z < 2.36) - P(0 < Z < 0.94) \\ = 0.4909 - 0.3264 = 0.1645$$

ERRATA 6: the slide PRINTS $Z2 = -0.4$, which is WRONG.
The correct value is -0.94 .
The slide's final 0.1644 is also wrong: $0.4909 - 0.3264$ equals 0.1645 using rounded table areas. Using unrounded z values gives about 0.1637 . WRITE $Z2 = -0.94$ and state which precision convention you used.

10. WORKED: LED bulbs (p015-p016)

"heavily skewed", $\mu = 50000$, $\sigma = 8000$, $n = 64$.
 $P(\bar{X} < 48000)$?

$$SE = 8000/\sqrt{64} = 8000/8 = 1000$$

$$Z = (48000 - 50000)/1000 = -2.00$$

$$P = 0.0228$$

note the SAME $-2.00 \rightarrow 0.0228$ as the normal example.
It is the same table read, just reached through SE instead of σ .

11. WORKED: machine life, small n (p017-p018)

"the lives follow the NORMAL distribution", $\mu = 7$, $\sigma = 1$, $n = 9$.
 $P(6.4 < \bar{X} < 7.2)$?

$$SE = 1/\sqrt{9} = 1/3$$

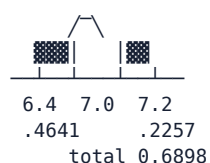
$$Z1 = (6.4 - 7)/(1/3) = -1.80$$

$$Z2 = (7.2 - 7)/(1/3) = 0.60$$

$$P = 0.4641 + 0.2257 = 0.6898$$

$n = 9$ IS BELOW 30, BUT THE QUESTION SAYS THE POPULATION IS NORMAL. That is case 2: the sample mean is EXACTLY normal. So we can still use the table. The word "normal" in the question gives an exact result, so no $n \geq 30$ heuristic is needed.

the two tails added:



THE SAMPLING/CLT KIT

SE of a mean	$\sigma/\sqrt{n}$ (or $s/\sqrt{n}$)
SE of a proportion	$\sqrt{p(1-p)/n}$
Z for a mean	$(\bar{X} - \mu)/(\sigma/\sqrt{n})$
Z for a single obs	$(X - \mu)/\sigma$ <- NOT the same!
$n \geq 30$	course heuristic for CLT approx
$n < 30$, normal pop	exact normal
$n < 30$, non-normal	normal approx needs justification

THE SINGLE COMMONEST MISTAKE IN THIS TOPIC:
using σ in the denominator for a question about the AVERAGE
of n items. An average ALWAYS divides by $\sigma/\sqrt{n}$.
one bulb's life -> divide by σ
the MEAN of 25 bulbs -> divide by $\sigma/\sqrt{25} = \sigma/5$

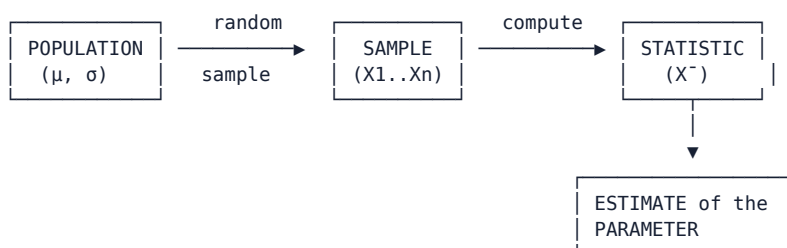
09 THEORY OF ESTIMATION (ppt5, 26 pages, lectures 19-21)

1. The vocabulary (p003-p004)

PARAMETER	a numerical characteristic of a POPULATION μ, σ^2, p
STATISTIC	a number computed FROM SAMPLE DATA $\bar{X}, S^2, \hat{p}$
ESTIMATOR	a FUNCTION of the sample used to estimate a parameter. It is a RANDOM VARIABLE. e.g. $\bar{X}$ as an estimator of μ
ESTIMATE	the NUMBER you get once data is plugged in e.g. $\bar{x} = 52$
PARAMETER SPACE	all possible values of the parameter $\theta = (-\infty, \infty)$ for μ

ESTIMATOR vs ESTIMATE - one mark, easy to lose
 $\bar{X}$ = the RULE (an estimator, a random variable)
 $\bar{x} = 205$ = the NUMBER (the estimate, once data is in)

2. The estimation pipeline (p004)



3. Point vs interval estimation (p005-p007)

POINT ESTIMATION

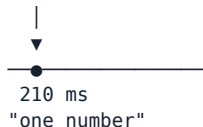
a SINGLE value

$$\mu \approx \bar{X} \quad \sigma^2 \approx S^2 \quad p \approx \hat{p}$$

- ✓ simple, easy to report
- ✗ does not show uncertainty

the picture from p007:

POINT



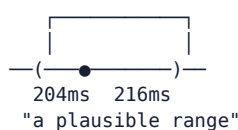
INTERVAL ESTIMATION

a RANGE of values

with a stated confidence level

- ✓ accounts for uncertainty
- ✓ more informative

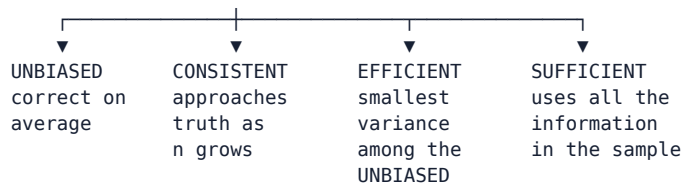
INTERVAL



$$CI = \text{Point Estimate} \pm \text{Margin of Error}$$

4. The four characteristics of a good estimator (p008)

A GOOD ESTIMATOR

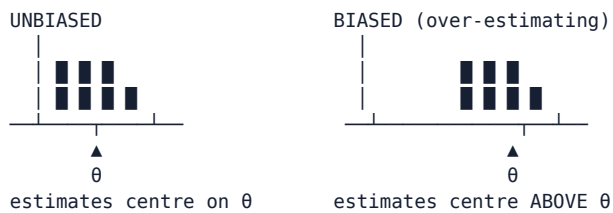


"good = little systematic error + stability + precision"

5. Unbiasedness (p009)

$\hat{\theta}$ is UNBIASED for θ if $E(\hat{\theta}) = \theta$
 $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$

the picture: many samples, plot the estimates



"CORRECT ON AVERAGE, NOT PER SAMPLE"
 An unbiased estimator can be badly wrong on any ONE sample.
 Unbiasedness is a statement about the LONG RUN.

6. Consistency (p010-p012)

$\hat{\theta}_n$ converges in probability to θ as $n \rightarrow \infty$
 $\hat{\theta}_n \xrightarrow{P} \theta$

i.e. for any $\epsilon > 0$: $P(|\hat{\theta}_n - \theta| < \epsilon) \rightarrow 1$ as $n \rightarrow \infty$

SUFFICIENT CONDITIONS FOR CONSISTENCY (a sequence T_n):

1. $E(T_n) \rightarrow \theta$ as $n \rightarrow \infty$
2. $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$

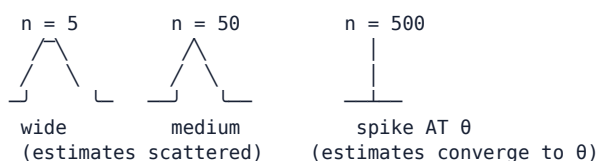
Both hold $\Rightarrow T_n$ is consistent.

WHY $\bar{X}$ IS CONSISTENT (p012) - the standard answer:

CONDITION 1	CONDITION 2	CONCLUSION
$E(\bar{X}) = \mu$ so $E(\bar{X}) \rightarrow \mu$	$\text{Var}(\bar{X}) = \sigma^2/n$ so $\sigma^2/n \rightarrow 0$	both hold $\Rightarrow \bar{X}$ is consistent

"more observations $\rightarrow$ smaller variance $\rightarrow$ more stable estimate"

the picture:



7. Efficiency (p013-p014)

$\hat{\theta}^1$ and $\hat{\theta}^2$ both UNBIASED for θ :
 if $\text{Var}(\hat{\theta}^1) < \text{Var}(\hat{\theta}^2)$ then $\hat{\theta}^1$ is MORE EFFICIENT

Among all UNBIASED estimators, the most efficient has the SMALLEST variance.
 Efficiency $\propto 1/\text{Variance}$

the slide's visual:

Estimator A
unbiased, Var = 4



concentrated
higher precision
-> MORE EFFICIENT

Estimator B
unbiased, Var = 9



spread out
lower precision
-> less efficient

THE ORDER OF OPERATIONS IN ANY COMPARISON:

- STEP 1 find which are UNBIASED (compute E of each)
- STEP 2 a biased one is ELIMINATED, even if its variance is smaller ("efficiency" only compares the unbiased)
- STEP 3 compare variances of the SURVIVORS, smallest wins

8. WORKED COMPARISON SET 1 (p015-p018) - the flagship question

Sample of size 5 from $N(\mu, \sigma^2)$. ($X_1..X_5$)

```
t1 = (X1+X2+X3+X4+X5)/5      <- the sample mean
t2 = (X1+X2)/2 + X3
t3 = (2X1 + X2 + λX3)/3      <- λ chosen for unbiasedness
```

STEP 1: FIND λ

$E(t3) = [2E(X1) + E(X2) + \lambda E(X3)]/3$
 $= (2\mu + \mu + \lambda\mu)/3 = [(3+\lambda)/3] \mu$
 for unbiasedness: $(3+\lambda)/3 = 1 \rightarrow \lambda = 0$
 so $t3 = (2X1 + X2)/3$

STEP 2: WHICH ARE UNBIASED

$E(t1) = (\mu+\mu+\mu+\mu+\mu)/5 = \mu$ ✓ unbiased
 $E(t2) = (\mu+\mu)/2 + \mu = 2\mu$ ✗ BIASED
 $E(t3) = (2\mu+\mu)/3 = \mu$ ✓ unbiased

STEP 3: COMPARE THE VARIANCES (of the unbiased ones)

$\text{Var}(t1) = \sigma^2/5 = 0.2000 \sigma^2$
 $\text{Var}(t3) = (4\sigma^2 + \sigma^2)/9 = 5\sigma^2/9 = 0.5556 \sigma^2$

ordering: $\sigma^2/5 < 5\sigma^2/9 < 3\sigma^2/2$
 (t1) (t3) (t2, biased)

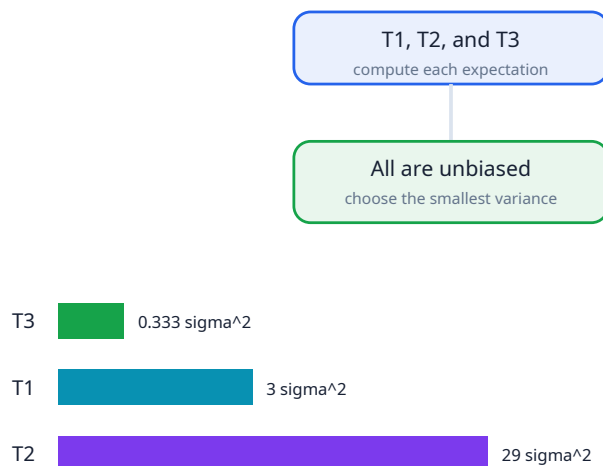
BEST = $t1 = \bar{X}$, the sample mean

the variance bar chart:

t1		0.200 σ^2	<- WINNER
t3		0.556 σ^2	
t2		1.500 σ^2	(biased, disqualified)

9. WORKED COMPARISON SET 2 (p019-p021) - with consistency

Estimator set 2: all unbiased, compare variance



Consistency needs a sequence T_n as n grows; fixed $n=3$ cannot prove it.

Unbiased estimators compared by variance

Sample of size 3. T_1, T_2, T_3 estimate μ .

$$T_1 = X_1 + X_2 - X_3$$

$$T_2 = 2X_1 + 3X_3 - 4X_2$$

$$T_3 = (\lambda X_1 + X_2 + X_3)/3$$

(i) ARE T_1 and T_2 UNBIASED?

$$E(T_1) = \mu + \mu - \mu = \mu \quad \checkmark$$

$$E(T_2) = 2\mu + 3\mu - 4\mu = \mu \quad \checkmark$$

BOTH unbiased

(ii) FIND λ FOR T_3

$$E(T_3) = (\lambda\mu + \mu + \mu)/3 = ((\lambda+2)/3)\mu$$

$$((\lambda+2)/3)\mu = \mu \rightarrow \lambda = 1$$

so $T_3 = (X_1 + X_2 + X_3)/3 = \bar{X}$, the sample mean

(iii) IS T_3 CONSISTENT?

With $\lambda=1$, this fixed- n statistic is the sample mean for $n=3$.

That fact alone does not establish consistency. The sequence $\bar{X}_n$ is consistent under the usual iid assumptions because $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$.

(iv) VARIANCES

$$\text{Var}(T_1) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) = 3\sigma^2$$

$$\text{Var}(T_2) = 4\text{Var}(X_1) + 9\text{Var}(X_3) + 16\text{Var}(X_2) = 29\sigma^2$$

$$\text{Var}(T_3) = (1/9)(3\sigma^2) = \sigma^2/3$$

minimum $\rightarrow T_3$ is the BEST estimator

the variance bar chart:



THE CONSISTENCY TRAP (worth memorising):
 T_1 's variance is $3\sigma^2$ for a FIXED $n=3$.
 That is NOT a consistency statement.
 Consistency is about behaviour AS n GROWS.
 A fixed- n variance, however small or large, says nothing about consistency.

10. Sufficiency (the lms-theory extra)

an estimator is SUFFICIENT if it uses ALL the information about θ present in the sample
the test is the NEYMAN-FISHER factorisation

examples in the corpus: the Poisson sample sum, the exponential sample mean, the sufficient statistics MCQ on MTE 2025-26 Q3

11. WORKED Real-Life 1: response time (p019)

response times of 8 requests: 180,210,195,220,205,190,200,240

$$\bar{x} = (\text{sum})/8 = 1640/8 = 205 \text{ ms}$$

ESTIMATOR = $\bar{x}$ (the rule)
ESTIMATE = 205 ms (the number)

THE QUESTION ASKS FOR **BOTH** "the estimator and the estimate". Two marks. Do not give only the number.

12. WORKED Real-Life 2: packet success (p020)

500 packets, 465 delivered. Estimate p .

$$\hat{p} = X/n = 465/500 = 0.93 = 93\%$$

estimator used: $\hat{p}$

13. WORKED Real-Life 3: battery life (p021) - the CI boundary item

$n = 25$, $\bar{x} = 8.4$ hours, $\sigma = 1.5$ known. 95% CI, $z(0.975) = 1.96$

This z interval assumes a normal population or a separately justified normal approximation. A sample size of 25 alone does not supply that justification.

$$\begin{aligned} \text{CI} &= \bar{x} \pm z(\sigma/\sqrt{n}) \\ &= 8.4 \pm 1.96(1.5/5) \\ &= 8.4 \pm 1.96(0.3) \\ &= 8.4 \pm 0.588 \\ &= (7.81, 8.99) \text{ hours} \end{aligned}$$

the picture:



NOTE: the CI MECHANICS belong to lecture 25 (BOUNDARY for this audit), but "point and interval estimation" IS named in the MTE syllabus line 15. Keep the formula in reserve; do not drill the mechanics.

14. The comparison quick table (p022)

#	PARAMETER	STATISTIC	RESULT
1	μ	$\bar{x}$	205 ms
2	p	$\hat{p}$	0.93
3	μ	$\bar{x}$	95% CI (7.81,8.99)

point -> one value

interval -> a range

15. The standard unbiased results (the toolkit)

ESTIMATOR	ESTIMATES	VARIANCE
$\bar{X}$	μ	σ^2/n
$S^2 = \sum (X_i - \bar{X})^2 / (n-1)$	σ^2	(the n-1 matters)
$\hat{p} = X/n$	p	$p(1-p)/n$

WHY n-1 AND NOT n IN S^2

The deviations are measured from $\bar{X}$, which ALREADY used the data. Using n would make S^2 too small on average. Dividing by n-1 corrects that. It is the price of estimating the mean from the same data.

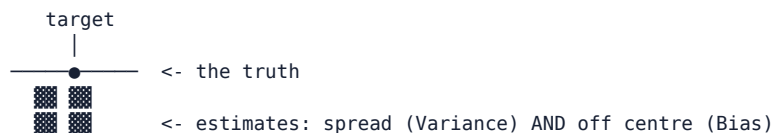
16. MSE, the honest comparison (formula sheet addition)

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

the picture:

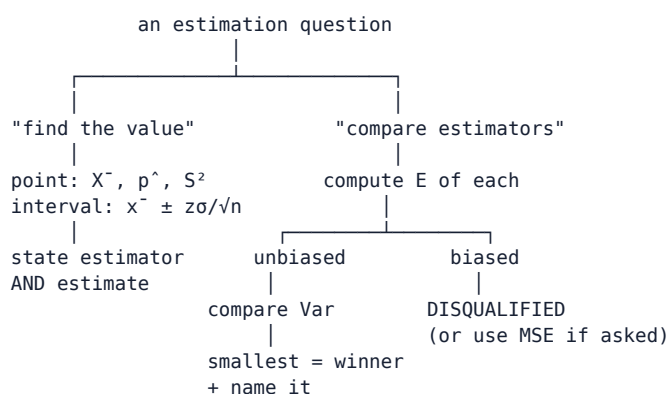
$$\text{MSE} = \text{VARIANCE} + \text{BIAS}^2$$

$\downarrow$ spread $\downarrow$ off-target



Efficiency ONLY compares unbiased estimators.
MSE lets a biased estimator compete, by paying for its bias in the second term. The exam could introduce this.

THE ESTIMATION DECISION TREE (for any question in this block)



That is the theory of estimation. The estimator-comparison layout (unbiased -> variances -> name the winner) is the 6-to-8 mark question of this block.

10 CHEBYSHEV AND THE HIDDEN LAYER (lectures 10-11 + H1-H8)

PART A: CHEBYSHEV'S INEQUALITY

1. The two forms, and why they exist

The theorem slide states BOTH:

$$\begin{aligned} \square P(|X - \mu| \geq k) &\leq \sigma^2/k^2 && \text{(the definition, } k \text{ in the EVENT)} \\ \square P(|X - \mu| \geq k \cdot \sigma) &\leq 1/k^2 && \text{(the } k\text{-sigma restatement)} \end{aligned}$$

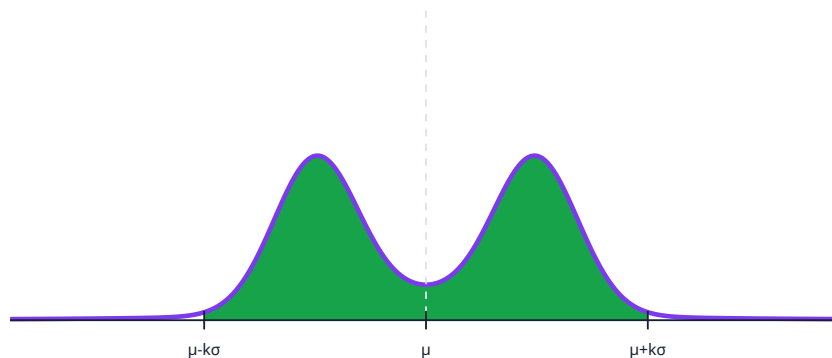
and the complement form:

$$\square P(|X - \mu| < k \cdot \sigma) \geq 1 - 1/k^2 \quad \text{(for "at least"/"within")}$$

CHEBYSHEV NEEDS ONLY μ AND σ . NO DISTRIBUTION IS ASSUMED.
It is always an UPPER BOUND on a TAIL.

2. The geometry, drawn

Chebyshev: guaranteed mass around the mean



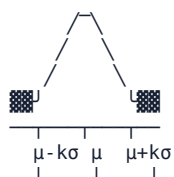
$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$$

valid for any distribution with finite mean and variance

schematic density, not a normal-distribution assumption

Chebyshev central-mass guarantee without a shape assumption

the event $|X - \mu| \geq k\sigma$ is the union of TWO tails:



each tail is part of the event
Chebyshev says: total tail $\leq 1/k^2$

and the complement $|X - \mu| < k\sigma$ is the CENTRE:



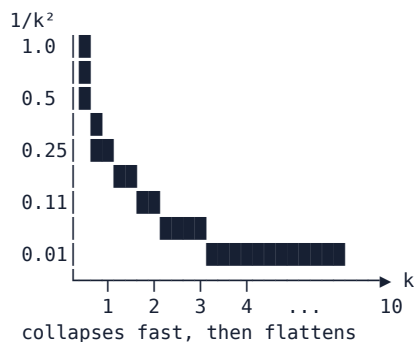


Chebyshev says: centre $\geq 1 - 1/k^2$

3. The thresholds table (memorise these)

k	$1/k^2$ tail	$1 - 1/k^2$ centre
1	1.0000	0.0000 (vacuous)
2	0.2500	0.7500 = 3/4
3	0.1111	0.8889 = 8/9
4	0.0625	0.9375 = 15/16
5	0.0400	0.9600 = 24/25
10	0.0100	0.9900 = 99/100

the shape of the bound:



4. The five facts the paper probes (formula sheet section E)

1. only a finite mean and variance are needed, NO distribution assumed
2. $k \leq 1$ gives a vacuous bound (says only " ≤ 1 ")
3. the bound is an UPPER bound on the tail, so a known-distribution question wants BOTH the bound and the exact value
4. k is measured in STANDARD DEVIATIONS, so convert $k = \varepsilon/\sigma$ first
5. the exact useful values: $k=2 \rightarrow 3/4$, $k=3 \rightarrow 8/9$, $k=4 \rightarrow 15/16$

5. WORKED: the corpus questions

— Q1 (safe) —

X has $E(X) = 3$ and $E(X^2) = 13$. Find a lower bound for $P(-2 < X < 8)$.

$$\text{Var} = E(X^2) - [E(X)]^2 = 13 - 9 = 4, \quad \sigma = 2$$

$$-2 < X < 8 \text{ is } |X-3| < 5, \text{ so } k = 5/2 = 2.5$$

$$P(|X-3| \geq 5) \leq 4/25$$

$$\text{complement } P(-2 < X < 8) = P(|X-3| < 5) \geq 21/25$$

— Q2 (safe) —

$\mu = 10$, $\sigma^2 = 4$ (so $\sigma = 2$)

$$P(|X - 10| \geq 3): \quad k = 3/2 = 1.5$$

$$\text{bound} = 1/k^2 = 1/2.25 = 0.4444 = 4/9$$

$$\text{complement } P(|X-10| < 3) \geq 1 - 4/9 = 5/9 = 0.5556$$

and the C=10 form: find c so $P(|X-10| \geq c) \leq 0.04$

$$1/k^2 = 0.04 \rightarrow k^2 = 25 \rightarrow k = 5$$

$$c = k \cdot \sigma = 5 \times 2 = 10$$

ERRATA 15: the deck's Q3 slide is a MESS.

- it STATES the row $(-1, -1, 3, 5)$ with $p = 1/6, 1/6, 1/6, 1/2$
- it WORKS the row $(-1, +1, 3, 5)$ (the second value +1)
- it writes a reciprocal template $P(|x-\mu| \geq k) < k^2/\sigma^2$

USE Q1 AND Q2 ONLY. Do not quote Q3's row.

(For the record, the WORKED row gives $E=3$, $E(X^2)=43/3$, $\text{Var}=16/3$, bound $16/3$. The STATED row would give $E=8/3$,

Var=65/9. They are different distributions.)

— WORKED: two dice (A1 short Q5) —
 Var(one die) = 35/12
 Var(sum of two) = 35/12 + 35/12 = 35/6 <- INDEPENDENCE RULE

$P(|X-7| \geq 3)$: $k = 3$, $\sigma^2 = 35/6$
 bound = $1/k^2 = 1/9$... no wait, use form $\square$:
 bound = $\sigma^2/k^2 = (35/6)/9 = 35/54 = 0.6481$

exact: sums ≤ 4 or ≥ 10 are 12 of 36 outcomes $\rightarrow 1/3 = 0.3333$

bound holds ($0.648 \geq 0.333$) and is much wider, as expected

— WORKED: 600 throws (A1 short Q6) —
 $X \sim B(600, 1/6)$: $\mu = 100$, $\sigma^2 = npq = 600(1/6)(5/6) = 250/3$

$P(80 \leq X \leq 120) = P(|X-100| \leq 20)$
 $\epsilon = 20$, $k^2 = \epsilon^2/\sigma^2 = 400/(250/3) = 1200/250 = 4.8$
 Chebyshev directly bounds the strict complement $|X-100| > 20$
 by at most 5/24, so the inclusive event is at least 19/24.

exact binomial = 0.9754

— WORKED: marks (A1 app Q4) —
 $\mu = 70$, variance = 25 ($\sigma = 5$). $P(60 < X < 80)$?
 $|X-70| < 10$, $k = 10/5 = 2$
 $P \geq 1 - 1/4 = 0.75$

6. The k-extraction recipe

1. get μ and σ ($\sigma = \sqrt{\text{Var}}$, compute Var if not given)
2. write the event as $|X - \mu| \geq \epsilon$ or $< \epsilon$
3. $k = \epsilon / \sigma$ <- RAW UNITS TO SIGMA UNITS
4. tail form $1/k^2$, or complement $1 - 1/k^2$

PART B: THE HIDDEN LAYER (H1-H8)

These carry marks and have NO teaching slide. Full treatment in [../11-BEYOND-SLIDES.md](#).

H1 INDEPENDENCE RULES OF RVs
 $E(XY) = E(X)E(Y)$ [needs independence]
 $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ [needs independence]
 $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ [the minus STAYS PLUS]
 $E(X+Y) = E(X) + E(Y)$ [ALWAYS, no condition]
 gates: dice-sum variance, $\text{Var}(X^+)$, estimator variances,
 Poisson additivity, every CLT SE

H2 CHEBYSHEV (the whole topic) - taught only in the L10-11
 deck, which arrived 15 Sep. Graded 9+ times.

H3 MEMORYLESS $P(T>s+t | T>s) = P(T>t)$
 exponential never ages. $P(T>t) = e^{(-\lambda t)}$.
 graded: A2 MCQ Q5, A2 section C Q1.

H4 NORMAL LANDMARKS 68.27 / 95.45 / 99.73 within 1,2,3 sigma
 graded: A2 MCQ Q11.

H5 EXPECTED COUNT = $N \times P$
 "out of N items, how many do we expect"

graded: A1 app, A2 B4 (5000 batteries), A2 D1 (10000 bulbs),
the 0.5499 die question, ETE family.
TRAP: report $N \times P$, not P .

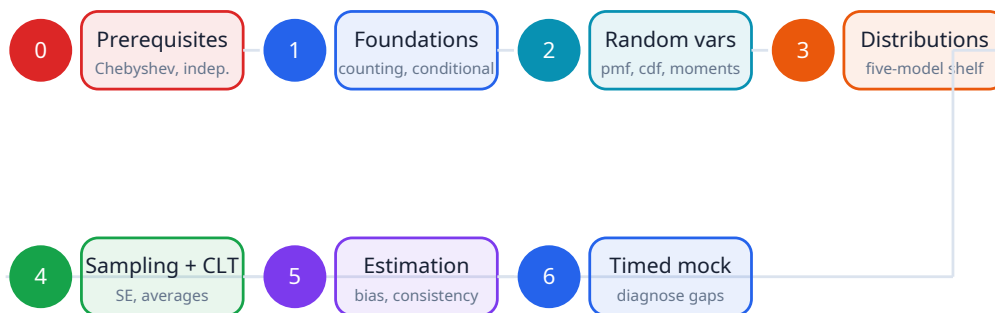
H6 GEOMETRIC $E(X) = 1/p$ (counts trials to first success)
graded: A1 short Q4 ($E = 2$).

H7 HYPERGEOMETRIC MEAN $= nK/N$ (draws WITHOUT replacement)
graded: A1 long Q3 ($4 \times 5/25 = 0.80$ exactly, both cases)
note it EQUALS the binomial mean np at the same p .

H8 THE TWO NORMAL TABLE CONVENTIONS $\phi(z)$ vs $F(z)$
 $\phi(z)$ = area from 0 to z ; $F(z)$ = cumulative from $-\infty$
differ by 0.5 for $z > 0$. DECIDE FIRST.

The hidden-layer study order and cost

Study order with dependency gates



advance only after the gate for the current pass is met

Study order and dependency gates

priority	item	why	time
1	H2	highest mark count (9+), gates nothing but is itself a whole topic	30m
2	H1	gates 4 downstream topics	25m
3	H8	every normal/CLT numeric	15m
4	H3	2 graded uses, one line to state	10m
5	H5	graded repeatedly, one idea	10m
6	H6	1 graded use	5m
7	H7	1 graded use	5m
8	H4	1 graded use, pure recall	5m

TOTAL $\approx$ 1h45

WHY THIS BLOCK IS THE BEST-SPENT TIME BEFORE THE PAPER:
it is the part of the syllabus that the slides UNDER-TAUGHT
but the papers KEEP ASKING. Cheapest marks available.